



Machine Design

Helicopter main rotor shaft (mast) - COURSE PROJECT

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Abstract

Helicopter mast plays a critical and essential role, ensuring the correct transfer of the loads and vibrations generated by the blades to the airframe. Therefore, a structural analysis aimed at accurately estimating the stress state of the component is necessary to verify that the structure is able to withstand the ultimate static load condition and fatigue loads. Given the geometric complexity of the component, the mast static analysis was performed shifting the focus to a simplified mast before, and to a specimen representative of the same stress state later. The analytical calculations performed on the specimen considering two materials, both for static and fatigue loads, suggest the safety of the component. Von Mises criteria was employed for static load assessment of the specimen, yielding appropriate safety factors of 1.55 and 1.48, while Sines multiaxial fatigue criteria was used for the fatigue assessment, suggesting safety factors of 1.71 and 1.51. Then, a numerical analysis was also carried out on the specimen considering the static loads and the maximum load condition for fatigue. The results endorse the safety of the specimen (and consequently of the real mast) by highlighting a good fit between analytical models and numerical analysis. Eventually, an experimental campaign for fatigue tests was carried out on the specimen, bolstering the outcomes of the finite element analysis, therefore validating the safety of the actual mast.

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1 Introduction

This report, commissioned by the former AgustaWestland S.p.A. (now Leonardo S.p.A.) concerns the study and structural validation of the main rotor shaft of the AW149 helicopter, a model specifically designed for the military sector. The main rotor shaft of an helicopter, also known as the main rotor mast, is an integral part of the main rotor system. This mechanical system is the most essential and safety-critical part of an helicopter, allowing the aircraft to generate lift and control authority. As suggested by Federal Aviation Administration (2023), the main rotor system can be broken down by identifying 3 main components: the blades, the hub, and the mast. At the heart of this system lies the mast, a critical mechanical element that transmits power from the power transmission system to the rotor blades. The main rotor shaft serves as a structural and dynamic interface between the main gearbox and the rotor hub, allowing rotational motion and torque to be transferred from the powerplant to the rotor blades. The upper end of the shaft is attached to the main rotor hub, which houses the rotor blades and transmits rotational motion to them. The hub is therefore responsible for introducing different types of loads into the mast. These loads induce a state of stress in the mast that is highly variable over time due to the aerodynamic origin of the forces and the rotation of the mast itself. This characteristic, combined with the safety-critical role that the mast plays, requires designers to create a component with high torsional stiffness, fatigue resistance, and corrosion durability, while being as lightweight as possible to optimize flight performance and fuel efficiency.

To properly assess the safety and capability of the AW149 mast to withstand the worst static load scenario and fatigue loads, a complete FEM analysis of the real component has been performed in order to depict the most stressed zones and the corresponding state of stress. Nonetheless, in the aerospace industry numerical outputs like FEM analysis are often flanked by experimental test campaigns. With regard to this case study, this common practice would be justified mainly by two reasons. Firstly, it is necessary to prove the validity and accuracy of the model employed in the FEM analysis, and secondly, it is required to assess and validate the fatigue behavior preliminarily estimated through analytical calculations. However, because of the geometrical complexity of the real component, it is not convenient and feasible to carry out experiments on the actual mast geometry. Therefore, two new specimens with simplified geometry have previously been designed to accurately reproduce the same stress state and the stress gradient of the real component in the most stressed zones. The two specimens, identical in size, differ from each other in the high-performance steel alloy they are made of: in the first case 9310 VIM VAR was used while in the second case 32CDV13 was the choice. To effectively state that the specimen can be considered as a substitute for the real component, an analytical evaluation under static load conditions must be performed both on a simplified but still accurate geometry of the mast and on the specimen just designed. If the results of the two analyses feature a similar stress state level, it will be possible to pursue the objective of the present work, i.e. the evaluation of the stress state due to static and fatigue loads acting on the specimen. At the beginning, the analysis will be carried out focusing on the most stressed section and the corresponding stress state of a simplified mast geometry in static condition. Later, the same analysis will be performed on the specimen; once the results in terms of equivalent stresses have been compared, the fatigue analysis and the experimental campaign on the specimen will be executed.

1.1 State of the art

Today helicopters can afford to operate in environments and situations that were unthinkable until a few years ago: all this has been possible thanks to significant advancements in materials engineering, computational modeling, and fatigue analysis. For example, modern masts must withstand complex combinations of torsion, bending, axial loads, and vibratory stresses induced by rotor dynamics while maintaining structural integrity over a long service life. In order to withstand such load scenarios, masts are currently made of high-strength steel or titanium alloys, although composite materials are increasingly

being explored for their favorable strength-to-weight ratios and fatigue resistance. The progressive use of composite materials in the aerospace industry therefore is modifying and adapting safety criteria. In fact, it should be mentioned that the study of fatigue in composite materials is actually a present subject of research by scholars around the world. For instance, a study by Rasuo (2009) conducted full-scale fatigue testing on helicopter blades made from composite laminated materials, simulating inflight loading conditions to assess structural adequacy. The applied test loads included simulated vibratory chordwise and flapwise bending, and vibratory torsional pitch motion. Post-test analyses revealed no delamination or permanent deformation, demonstrating the effectiveness of the design and testing approach.

At the same time, innovative design philosophies, such as the damage-tolerance approach, are progressively arousing the interest of designers and structural engineers as well. This brand new approach, thanks to discoveries on fracture mechanics made in the previous century, ensures the safety and durability of a component even after it has sustained damage. Instead of assuming that materials are flawless, damage-tolerance approach acknowledges that small cracks or defects may exist and focuses on predicting how these flaws grow over time under normal usage. By understanding and monitoring crack growth, engineers can therefore design structures that remain safe until damage is detected and repaired, preventing sudden failures and improving maintenance planning. For instance, Everett and Elber (1999) presented at the NATO/AVT meeting on aging rotorcraft systems a study that applied the damage-tolerant design philosophy to helicopter components. The authors showed how a tail-rotor pitch rod can be successfully assessed with a damage-tolerant design philosophy. The study was also able to establish non-destructive inspection schedules.

Among modern technologies, the integrate health and usage monitoring systems (HUMS) takes a special place thanks to its capability to continuously assess aerospace structure operational loads and detect early signs of damage or fatigue accumulation in service. In fact, HUMS uses sensors to track the operational status and health of critical components, to predict and prevent potential failures. This allows designers to set condition-based maintenance strategies that enhance safety and reduce lifecycle costs.

Lastly, physical testing is nowadays a common practice for aerospace structures design. In fact, it allows to refine the design itself and the assessment methodologies, improving performance, safety, and reliability with respect to the past. For example, full-scale fatigue testing under variable amplitude loading, along with non-destructive inspection techniques like ultrasonic testing and eddy current inspection, are used to validate these life predictions and monitor for damage progression.

1.2 Methods used

Since the material that is dealt with is a ductile material it is possible to proceed with the assessment by considering two type of criterion which are the Von-Mises and Guest-Tresca.

The one chosen in this report to carry out the assessment is the Von-Mises which a powerful tool to predict when a material might start to yield under certain loading conditions, and as said before, it is particularly suitable for ductile material.

The Von-Mises stress helps engineers simplify the complex stress states that occur in materials since it combines the effects of different stress components into a single value, making it easier to compare against a material's yield strength.

This criterion, also known as the maximum distortion energy criterion, is based on defining the so called Von-Mises stress which can be derived from the state of the material taken in consideration.

The mathematical expression that is needed to be satisfied is the following one:

$$\sigma_{Vm} \leq \frac{\sigma_{yield}}{\eta}$$

The equivalent stress can be calculated in many way, for example by exploiting the terms of the stress components in the x, y, and z directions and the shear stresses or using the principal stresses; the latter is the one used in this report.

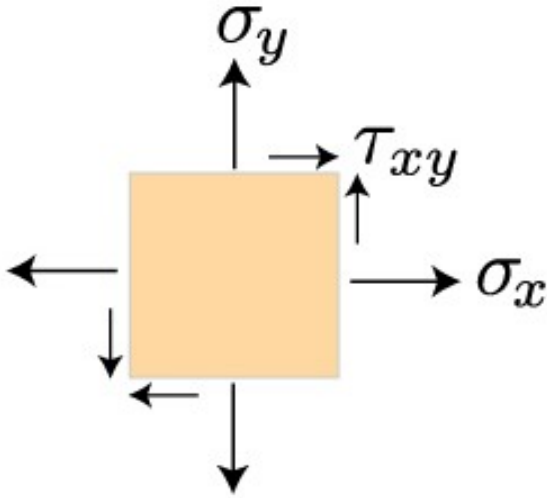


Figure 1.1: Generic plane state of stress

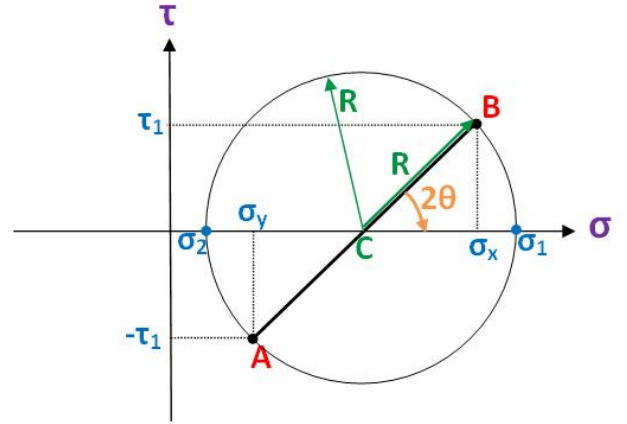


Figure 1.2: Generic Mohr circle

In particular the principal stresses are the normal stresses that occur at particular orientations where the shear stress is zero, they consist in the eigenvalues of the stress tensor and represent the maximum and minimum normal stresses that a material element experiences.

They are crucial for understanding the state of stress at a certain point in the material.

To calculate the principal stress, it is possible to proceed by identifying the state of stress of the component, which in our application, is going to be a 2D state of stress (plain stress) like the one in figure 1.1.

Then it is possible to study two point of this state of stress where each one can be identified with two coordinates that are the corresponding normal stress σ and shear stress τ of that side of the state of stress.

This is useful to define the Mohr circle (figure 1.2) in the $\sigma - \tau$ plane, which is a graphical representation of the state of stress at a point in a material helpful to provide a visual method to determine principal stresses.

After calculating the radius and the coordinate of the center it is possible to calculate the value of the principal stress which are the points where the Mohr circle intersects the σ axis and since the state of stress is 2D, one of the 3 principal stress is going to be null.

The following step is the one where the equivalent Von-Mises stress can be calculated:

$$\sigma_{Vm} = \sqrt{\sigma_I^2 + \sigma_{II}^2 + \sigma_{III}^2 - \sigma_I\sigma_{II} - \sigma_{II}\sigma_{III} - \sigma_{III}\sigma_I}$$

In conclusion, the Von-Mises criterion is a tool that engineers can use to ensure the safety and reliability of structures and mechanical components, since it helps in designing components that are going to withstand operational loads without undergoing permanent deformation. Also it is a simplified approach to proceed with the assessment of complex stress states allowing good accuracy regarding when the yielding of the material will happen.

1.3 Specimen and mast material properties

Property	Symbol [Unit]	9310 VIM VAR
Young's modulus	E [MPa]	200000
Shear modulus	G [MPa]	76000
Poisson's ratio	ν	0.32
Yielding strength	S_{ys} [MPa]	893
Ultimate tensile strength	U_{ts} [MPa]	1020
Endurance limit	$\sigma_{FA,a}$ [MPa]	350

Table 1.1: Mechanical properties of Material 1: 9310 VIM VAR

Property	Symbol [Unit]	32CDV13
Young's modulus	E [MPa]	200000
Shear modulus	G [MPa]	76000
Poisson's ratio	ν	0.30
Yielding strength	S_{ys} [MPa]	850
Ultimate tensile strength	U_{ts} [MPa]	1000
Endurance limit	$\sigma_{FA,a}$ [MPa]	320

Table 1.2: Mechanical properties of Material 2: 32CDV13

2 Static assessment mast

2.1 Section geometry (mast)

Parameter	Value	Description
R_i	70 mm	Internal radius (critical section)
t	9 mm	Shaft thickness (critical section)
l_1	438 mm	Distance from boundary condition 1 (b.c.1)
l_2	202 mm	Distance from load application point
$K_{t,a}$	1.12	Notch coefficient under tension
$K_{t,t}$	1	Notch coefficient under torsion

Table 2.1: Section geometry

2.2 Loads on the shaft (mast)

It is needed to analyze the mast in a simplified way, thus we can consider it as a hollow shaft.

Considering that the mast is fixed at one end, it can be seen as supported by a pin (at the fixed part) and a roller where a notch, in particular a shoulder, is present.

On the other end of the shaft the loads are applied, which consist of the force applied by aerodynamics and by the torque of the engine-transmission block. A scheme of loads applied to the mast and reaction forces given by constraints is shown in figure 2.1

Thus, it is possible to look at the mast as a shaft with the following:

Constraint reaction given by pin:

- V_1
- H_1
- Cm

Constraint reaction given by the roller:

- V_2

Loads defined by the working condition:

$$N = 1000 \text{ N} \quad (2.1)$$

$$T = 32000 \text{ N} \quad (2.2)$$

$$M_f = 35000 \text{ Nm} \quad (2.3)$$

$$M_t = -80000 \text{ Nm} \quad (2.4)$$

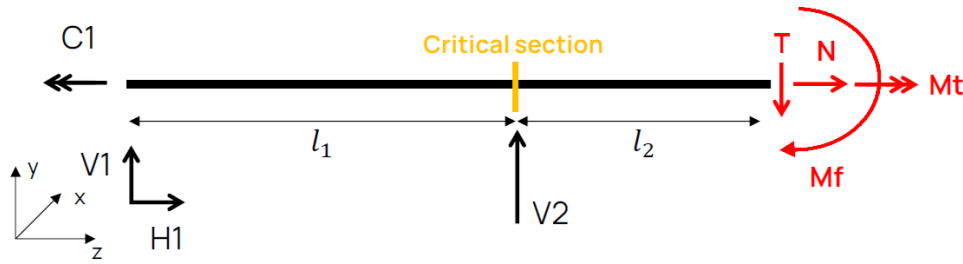


Figure 2.1: Mast loads and reaction forces

2.3 Boundary Conditions and force distribution

2.3.1 Boundary Conditions reactions

By applying the following system to solve the structure:

$$\begin{cases} \sum_{i=1}^n F_y = 0 \\ \sum_{i=1}^n F_z = 0 \\ \sum_{i=1}^n M_{fz} = 0 \\ \sum_{i=1}^n M_{tx} = 0 \end{cases}$$

The following results are obtained:

$$\begin{cases} V_1 = -94666.7N \\ H_1 = -100000N \\ V_2 = 12666.7N \\ Cm = -80000Nm \end{cases}$$

2.3.2 Diagrams

The internal forces diagrams are shown in figure 2.2:

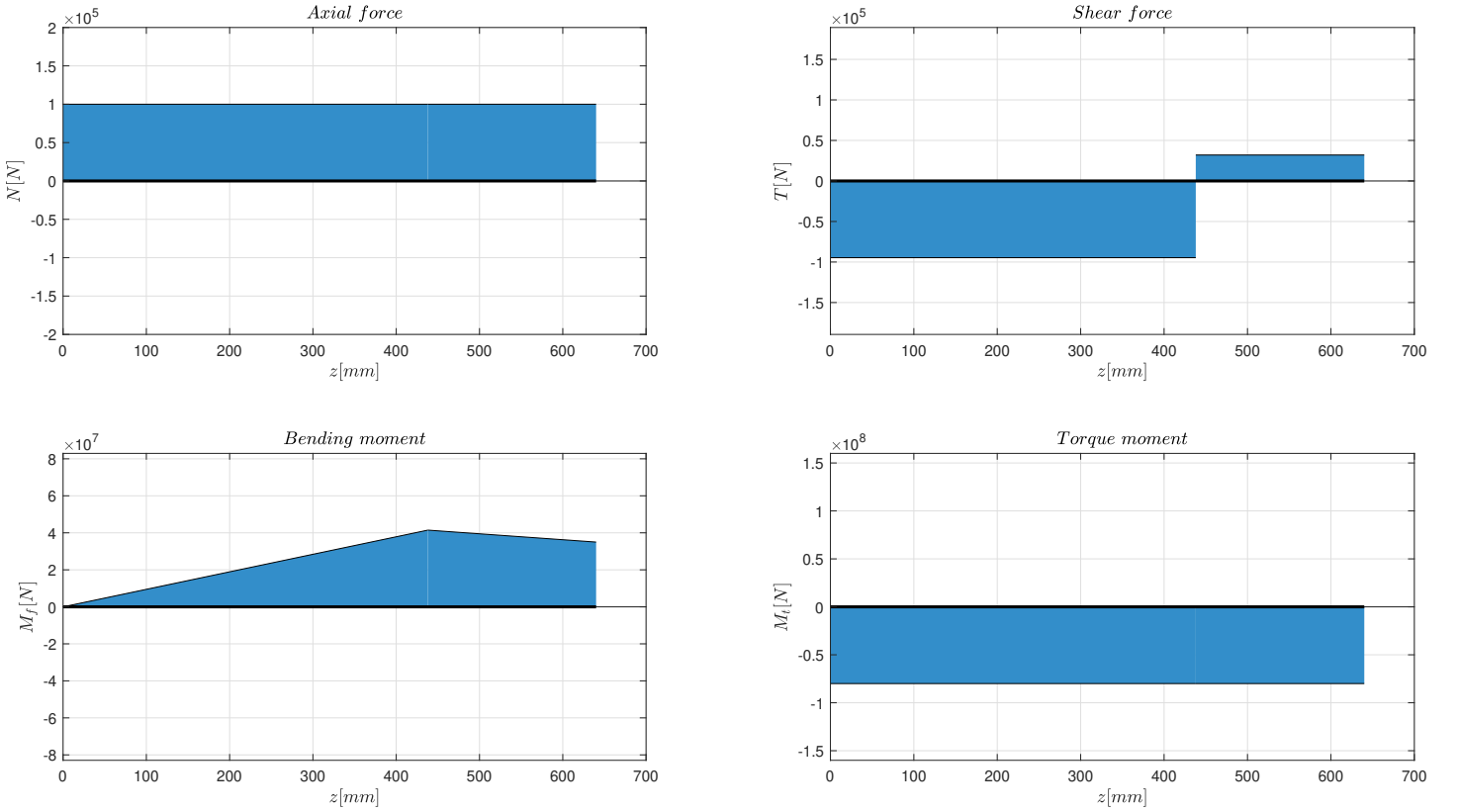


Figure 2.2: Mast internal forces diagrams

2.4 Most stressed point of the most stressed section

By analyzing the internal force diagrams and accounting for the presence of a notch, it is evident that the most critically stressed section corresponds to the location where the roller support is applied.

Furthermore, since the shear force is relatively low, the shear stress τ , which follows a parabolic distribution, is also comparatively small. In contrast, the stresses induced by the bending moment and torsional moment are significantly higher. As a result, the most stressed points are located along the line perpendicular to the neutral axis of the bending moment, particularly on the outer surface of the shaft, where the shear stress due to the torsional moment reaches its maximum.

Although the shear stresses resulting from the transverse shear force exhibit a parabolic distribution, with their peak occurring at a different point, this location is not the most critically stressed. This is because, at that point, no axial stress due to bending is present, and thus the combined stress state is less severe.

2.4.1 State of stress

Given that the most stressed point is P_1 , the state of stress of this point will consist in axial stress given by the axial force together with bending momentum and a shear stress given by the torque momentum only.

Where the axial stress is:

$$\sigma = \frac{N}{\pi (R_e^2 - R_i^2)} + \frac{32 M_f D_e}{\pi (D_e - D_i)^4} = 302.90 MPa$$

And the shear stress is:

$$\tau = \frac{16 M_t D_e}{\pi (D_e - D_i)^4} = 269,30 MPa$$

The results are obtained by considering $R_e = 0,079m$ and $R_i = 0,070m$.

The effect of the notch produce a peak in the stress distribution, which considering that the $K_f = 1,12$ and $k_t = 1$, generates the following peaks in stress:

$$\sigma_{max} = K_f \sigma = 339,25 MPa$$

$$\tau_{max} = K_t \tau = 269,30 MPa$$

Considering an infinitesimal element of the material in the point selected, to study the state of stress of it, it is possible to consider two particular points A and B have the following coordinates:

$$A = (339,25 ; 269,30) \quad (2.5)$$

$$B = (0 ; -269,30) \quad (2.6)$$

Which can be used to build the Mohr circle in the $\sigma - \tau$ plane, and studying the circle it is possible to identify:

$$x_c = \frac{\sigma_A + \sigma_B}{2} = 169,71 MPa$$

$$y_c = 0$$

$$r = \sqrt{x_c^2 + \tau_B^2} = 318,31 MPa$$

Thus:

$$\sigma_I = x_c + r = 488,01 MPa$$

$$\sigma_{II} = 0 MPa$$

$$\sigma_{III} = x_c - r = -148,61 MPa$$

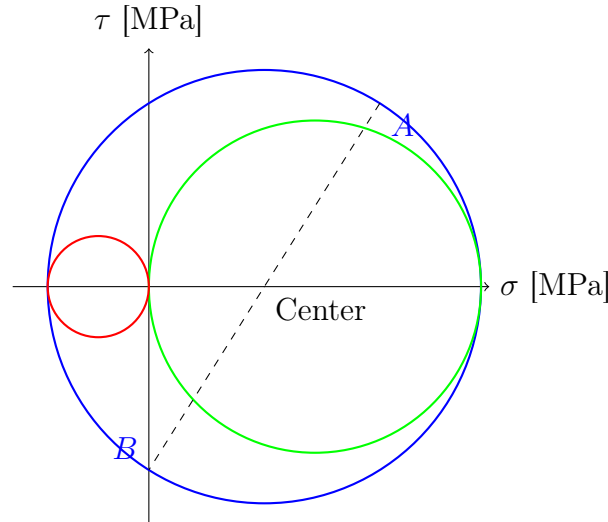


Figure 2.3: Mohr circle of the state of stress of the most stressed section during specimen static assessment

2.5 Verification of static resistance

What is left is the static assessment of the mast that can be done with the Von-Mises criterion:

$$\sigma_{Vm} = \sqrt{\sigma_I^2 + \sigma_{II}^2 + \sigma_{III}^2 - \sigma_I \sigma_{II} - \sigma_{II} \sigma_{III} - \sigma_{III} \sigma_I} = 576,77 MPa$$

The equivalent stress is lower than the yield stress, so the mechanical component can be considered safe under these loading conditions. Since we are talking about a ductile material, it is easy to compute the *Safety Factor* (η) as the ratio between the yield stress and the *Von Mises* one:

- for material 9310 VIMVAR $\eta = \frac{\sigma_{yield}}{\sigma_{Vm}} = 1,548$
- for material 32CDV13 $\eta = \frac{\sigma_{yield}}{\sigma_{Vm}} = 1,474$.

3 Static assessment specimen

3.1 Section geometry (specimen)

Parameter	Value	Description
R_i	20.55 mm	Internal radius (critical section)
t	2.92 mm	Shaft thickness (critical section)
l	460 mm	length of the specimen
$K_{t,a}$	1	Notch coefficient under tension
$K_{t,t}$	1	Notch coefficient under torsion

Table 3.1: Section geometry

3.2 Loads on the shaft (specimen)

In order to investigate the behavior of the component under realistic loading conditions, a specimen is used for laboratory testing, as the geometry and size of the actual mast make direct testing impractical.

This section presents the static analysis of the specimen using an analytical approach. A subsequent section will provide a comparison between the analytical results, experimental data, and numerical simulations performed using the Finite Element Method (FEM).

The laboratory setup consists of a testing machine designed with two primary components: a fixed support that clamps one end of the specimen to the ground and a loading mechanism at the opposite end. This system is able to apply at the extremity a combination of axial loads and torsional moments. However, for the purposes of the analytical study, bending moments will also be considered in order to more accurately replicate the real-world loading conditions experienced by the mechanical device.

The specimen can therefore be modeled as a hollow shaft, clamped at one end and subjected to axial force, torque, and bending moment at the other (3.1).

The specimen, free from the constrain (interlocking), appear with:

- A
- B
- C
- M_A
- M_B
- M_c

While on the other end the following loads are applied:

$$N = 29000 \text{ N} \quad (3.1)$$

$$M_f = 1019 \text{ Nm} \quad (3.2)$$

$$M_t = -2328 \text{ Nm} \quad (3.3)$$

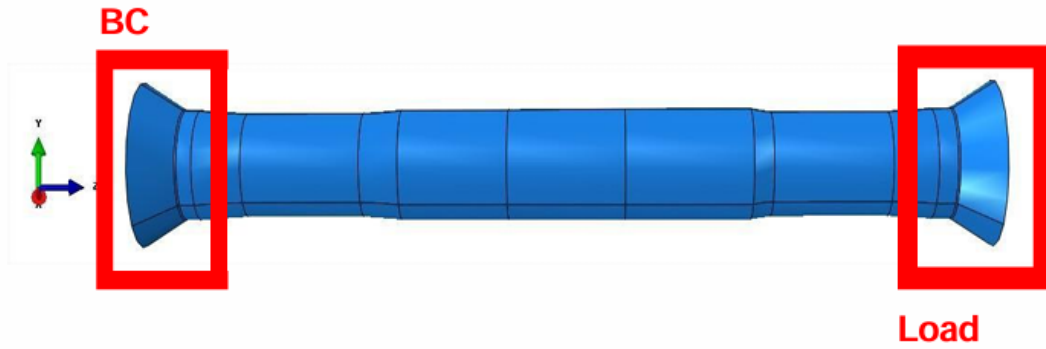


Figure 3.1: Specimen

3.3 Boundary Conditions and force distribution

3.3.1 Boundary Conditions reactions

The structure can be solved with this equilibrium equations:

$$\begin{cases} \sum_{i=1}^n F_x = 0 \\ \sum_{i=1}^n F_y = 0 \\ \sum_{i=1}^n F_z = 0 \\ \sum_{i=1}^n M_x^o = 0 \\ \sum_{i=1}^n M_y^o = 0 \\ \sum_{i=1}^n M_z^o = 0 \end{cases}$$

Which gives:

$$\begin{cases} A = 29000N \\ B = 0N \\ C = 0N \\ M_A = -2328Nm \\ M_B = 0Nm \\ M_C = 1019Nm \end{cases}$$

3.3.2 Diagrams

The diagrams of the internal forces for the static assessment of the specimen are shown in figure 3.2.

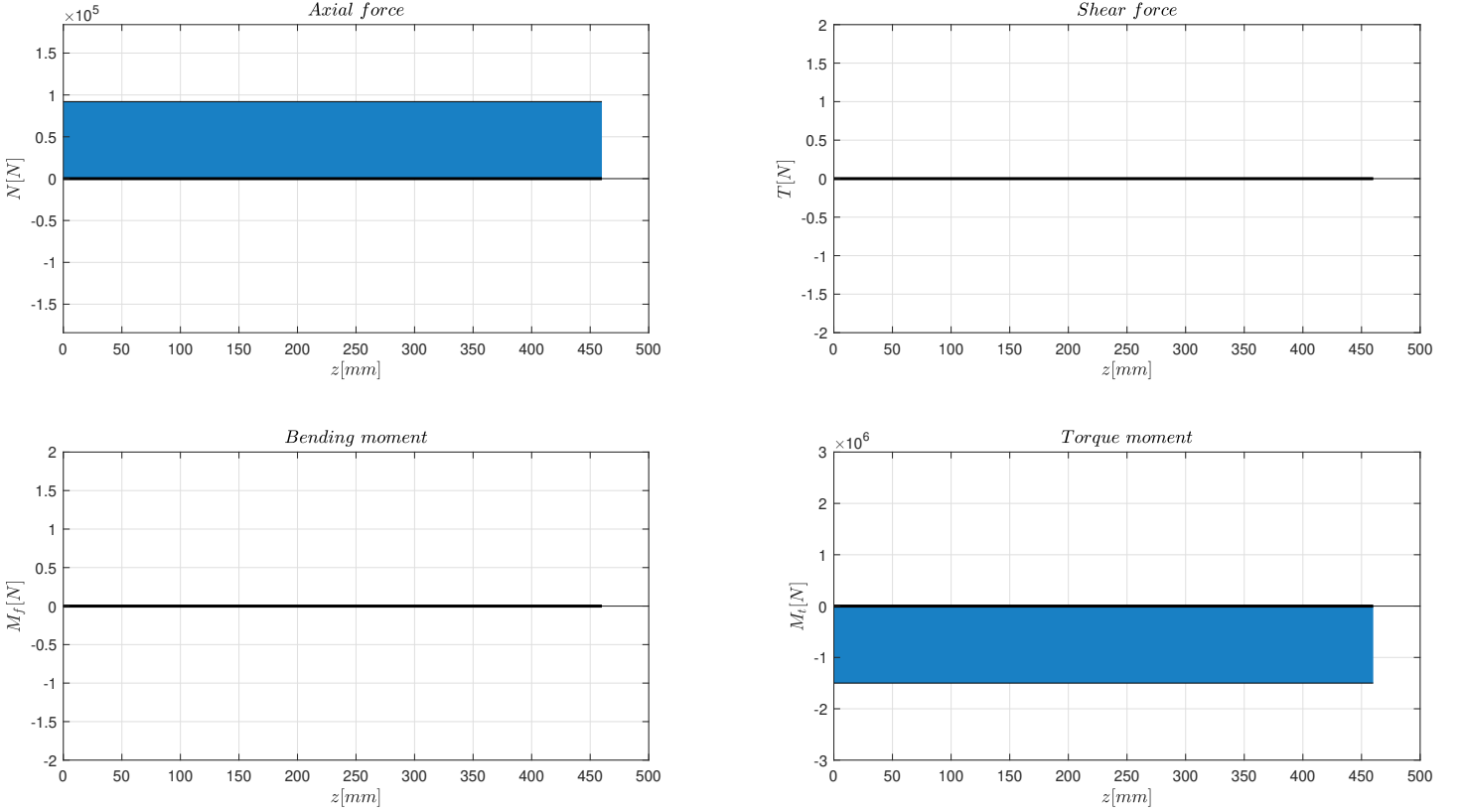


Figure 3.2: Specimen internal forces diagrams

3.4 Most stressed point of the most stressed section

By analyzing the diagrams it is possible to see that the forces and momentum are constant everywhere so a generic section can be considered, of course taking into account that a notch is on the specimen.

Since a bending momentum is acting on the specimen, the most stressed points will be on axis perpendicular to the neutral axis and in particular will be on the most external point where the axial stress and shear stress given by momentum will be at maximum.

3.4.1 State of stress

Given that the most stressed point is P_1 the state of stress of this point will consist in axial stress given by the axial force together with the bending momentum and a shear stress given by the torque momentum only.

Where the axial stress is:

$$\sigma = \frac{N}{\pi \left(\left(\frac{D_e}{2} \right)^2 - \left(\frac{D_i}{2} \right)^2 \right)} + \frac{32 M_f D_e}{\pi (D_e - D_i)^4} = 315,25 MPa$$

And the shear stress is:

$$\tau = \frac{16 M_t D_e}{\pi (D_e - D_i)^4} = 278,08 MPa$$

The results are obtained by considering as dimension of the specimen the one retrieved from the model which will be used in the FEM evaluation, which are $D_e = 0,04694m$ and $D_i = 0,0411m$ as external and internal diameter in the smallest section, $t = 0,00292m$ as thickness and $r = 0,07045m$ as radius of curvature of the shoulder.

Now it is time to take into consideration the notch, which has the effect of changing the stress distribution generating peaks in the value of the stresses.

In particular the value of $K_{tf} = 1$ and $K_{tt} = 1$ is used since the notch radius of curvature is really high with respect to the diameter of the specimen, so the shoulder step has a progressive transition regarding its geometry, meaning that the effect of the notch is negligible.

The peaks values are:

$$\sigma_{max} = K_{tf} \sigma = 315,25 \text{ MPa}$$

$$\tau_{max} = K_{tt} \tau = 278,08 \text{ MPa}$$

From the σ_{max} we can calculate what is the axial that need to be applied by the machine in the laboratory on the component (since it cannot apply a bending momentum) with the formula:

$$315,25 \text{ MPa} = \frac{F}{\pi \left(\left(\frac{D_e}{2} \right)^2 - \left(\frac{D_i}{2} \right)^2 \right)}$$

Which gives a axial force of $F \approx 127000 \text{ N}$.

Considering an infinitesimal element of the material in the point selected, the state of stress retrieved appears like this:

$$\boldsymbol{\sigma} = \begin{bmatrix} \sigma_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & \sigma_{yy} & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & \sigma_{zz} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 278,08 \text{ MPa} \\ 0 & 0 & 0 \\ 278,08 \text{ MPa} & 0 & 315,25 \text{ MPa} \end{bmatrix}$$

Now, the two particular points (the ones on sides of the infinitesimal element) A and B have the following coordinates:

$$A = (315,25; -278,08) \quad (3.4)$$

$$B = (0; 278,08) \quad (3.5)$$

Which can be used to build the Mohr circle in the $\sigma - \tau$ plain shown in figure 3.3.

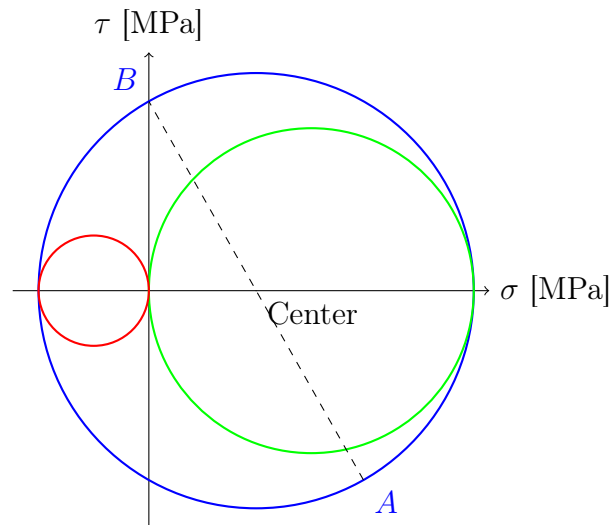


Figure 3.3: Mohr circle of the state of stress of the most stressed section during mast static assessment

3.4.2 Principal stresses in critical section

Studying the Mohr circle it is possible to identify:

$$x_c = \frac{\sigma_A + \sigma_B}{2} = 157,62 MPa$$

$$y_c = 0$$

$$r = \sqrt{x_c^2 + \tau_B^2} = 319,64 MPa$$

Thus:

$$\sigma_I = x_c + r = 477,27 MPa$$

$$\sigma_{II} = 0 MPa$$

$$\sigma_{III} = x_c - r = -162,02 MPa$$

So the principal stress tensor is

$$\boldsymbol{\sigma}_{principal} = \begin{bmatrix} 477,27 MPa & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -162,02 MPa \end{bmatrix}$$

3.5 Verification of static resistance

What is left is the static assessment of the specimen that can be done with the Von Mises criterion by adopting another formulation, which is the following:

$$\sigma_{VM} = \sqrt{\sigma_{max}^2 + 3\tau_{max}^2} = 575,93 MPa$$

The equivalent stress is lower than the yield stress, so the mechanical component can be considered safe under these loading conditions. Since we are talking about a ductile material, it is easy to compute the *Safety Factor* (η) as the ratio between the yield stress and the *Von Mises* one: Since the material taken into account is a ductile material:

- for material 9310 VIMVAR $\eta = \frac{\sigma_{yield}}{\sigma_{Vm}} = 1,551$
- for material 32CDV13 $\eta = \frac{\sigma_{yield}}{\sigma_{Vm}} = 1,477$

4 Fatigue assessment of the specimen

Now it's time to study the fatigue behavior of the specimen.

In this section, regarding the analytical part, the specimen will be again considered as a hollow shaft with an end fixed, which represent the interlocking of the machine in the laboratory, and the other end will be loaded.

Again it is advisable to remember the laboratory machine is not able to apply a bending moment thus the loads that are considered are the axial force and the torque moment.

So the free specimen will have again with the following reaction forces at the fixed end:

- H
- C

While on the opposite end the loads are applied with forces varying in time such as:

- $N = 12000 + 80000 \sin(\omega t) \text{ N}$
- $M_t = 1500Nm$

The specimen appears like this (Figure 4.1):

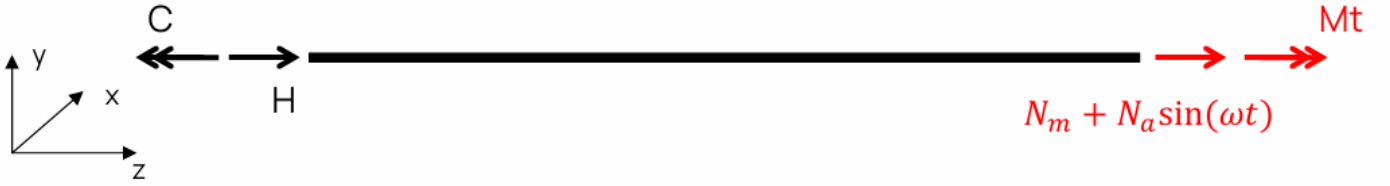


Figure 4.1: Schematization of the specimen under fatigue load

Of course by applying the equilibrium equation like:

$$\begin{cases} \sum_{i=1}^n F_z = 0 \\ \sum_{i=1}^n M_z^o = 0 \end{cases}$$

We obtain:

$$\begin{cases} H = -N \\ C = M_t \end{cases}$$

4.1 Stress cycle

It is possible to see, since only axial force and torque moment are applied, that the most stressed point is going to be on the surface of the specimen where the shear force is at maximum.

This loading condition creates respectively a axial stress and a shear stress.

Since the specimen is subjected to a fatigue loading condition, it is mandatory to consider the variation of the load, and considering that in this case only the axial load is a function of time we can identify the following value:

$$N_{max} = 12000N + 80000 = 92000N$$

$$N_{min} = 12000N - 80000 = -68000N$$

Again the formulas for the hollow circular section can be used to retrieve the maximum and minimum axial stress:

$$\sigma_{max} = \frac{N_{max}}{\pi \left(\left(\frac{D_e}{2} \right)^2 - \left(\frac{D_i}{2} \right)^2 \right)} = 227,9 MPa$$

$$\sigma_{min} = \frac{N_{min}}{\pi \left(\left(\frac{D_e}{2} \right)^2 - \left(\frac{D_i}{2} \right)^2 \right)} = -168,5 MPa$$

The results are obtained by considering again $D_e = 0,04694m$ and $D_i = 0,0411m$ as external and internal diameter in the smallest section.

And now it is time to calculate the amplitude and mean value of the axial stress:

$$\sigma_a = \frac{\sigma_{max} - \sigma_{min}}{2} = 198,11 MPa$$

$$\sigma_m = \frac{\sigma_{max} + \sigma_{min}}{2} = 29,72 MPa$$

While for the shear stress, considering the torque moment is constant, only a mean value can be calculated:

$$\tau_a = 0 MPa$$

$$\tau_m = \frac{16 M_t D_e}{\pi ((D_e)^4 - (D_i)^4)} = 179,17 MPa$$

4.1.1 Average stress tensor

Now it is time to focus just on the mean value of the stress, meaning that there is a contribute coming both from the axial force and torque moment.

With the state of stress taken in consideration it is possible to proceed by retrieving the mean stress tensor that is:

$$\boldsymbol{\sigma}_m = \begin{bmatrix} 0 & 0 & -\tau_m \\ 0 & 0 & 0 \\ -\tau_m & 0 & \sigma_m \end{bmatrix} =$$

$$= \begin{bmatrix} 0 & 0 & -179,17 MPa \\ 0 & 0 & 0 \\ -179,17 MPa & 0 & 29,72 MPa \end{bmatrix}$$

And studying the point A and B on the infinitesimal element with coordinate:

$$A = (29,72; 179,17) \quad (4.1)$$

$$B = (0; -179,17) \quad (4.2)$$

The Mohr circle for the mean stress can be retrieved and consequently the mean value of the principal stress:

$$x_c = \frac{\sigma_A + \sigma_B}{2} = 14,86 MPa$$

$$y_c = 0$$

$$r = \sqrt{x_c^2 + \tau_B^2} = 179,78 MPa$$

Thus:

$$\sigma_{Im} = x_c + r = 194,65 MPa$$

$$\sigma_{IIIm} = 0 MPa$$

$$\sigma_{IIIIm} = x_c - r = -164,93 MPa$$

So the mean principal stress tensor is:

$$\begin{aligned}\sigma_{\text{principal}_m} &= \begin{bmatrix} \sigma_{Im} & 0 & 0 \\ 0 & \sigma_{II_m} & 0 \\ 0 & 0 & \sigma_{III_m} \end{bmatrix} = \\ &= \begin{bmatrix} 194,65MPa & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -164,93MPa \end{bmatrix}\end{aligned}$$

4.1.2 Alternate stress tensor

A similar thing must be done for the alternate value of the stress, meaning that now there is a contribute coming only from the axial force.

With the state of stress taken in consideration it is possible to proceed by retrieving the alternate stress tensor that is:

$$\begin{aligned}\sigma_a &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \sigma_a \end{bmatrix} = \\ &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 198,11MPa \end{bmatrix}\end{aligned}$$

And studying the point A and B on the infinitesimal element with coordinate:

- $A = (198, 11; 0)$
- $B = (0; 0)$

It is possible to retrieve the Mohr circle for the alternate stress and consequently the mean value of the principal stress:

$$\begin{aligned}x_c &= \frac{\sigma_A + \sigma_B}{2} = 99,055MPa \\ y_c &= 0 \\ r &= x_c = 99,055MPa\end{aligned}$$

Thus:

$$\begin{aligned}\sigma_{Im} &= x_c + r = 198,11MPa \\ \sigma_{II_m} &= 0MPa \\ \sigma_{III_m} &= x_c - r = 0MPa\end{aligned}$$

So the mean principal stress tensor is:

$$\begin{aligned}\sigma_{\text{principal}_a} &= \begin{bmatrix} \sigma_{Ia} & 0 & 0 \\ 0 & \sigma_{IIa} & 0 \\ 0 & 0 & \sigma_{IIIa} \end{bmatrix} = \\ &= \begin{bmatrix} 198,11MPa & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}\end{aligned}$$

4.2 Principal component variation with time

The fatigue tests are characterized by a sinusoidal alternate axial load

$$N(t) = N_m + N_a \sin(\omega t) \quad (4.3)$$

and so it's possible to keep explicit the time dependence of the stress tensor

$$\boldsymbol{\sigma}(t) = \boldsymbol{\sigma}_m + \boldsymbol{\sigma}_a(t) = \begin{bmatrix} 0 & 0 & -179.17 \\ 0 & 0 & 0 \\ -179.17 & 0 & 29.72 + 198.11 \sin(\omega t) \end{bmatrix} MPa$$

Proceeding this way, it's immediate to compute the eigenvalues of the total stress tensor to find the principal stress components as function of time:

$$\det(\boldsymbol{\sigma}(t)) = 0 \quad (4.4)$$

$$\begin{aligned} \sigma_I &= \sigma_I(t) \\ \sigma_{II} &= \sigma_{II}(t) \\ \sigma_{III} &= \sigma_{III}(t) \end{aligned} \quad (4.5)$$

The resulting principal stress components are shown in figure 4.2 (where the angular frequency $\omega = 1 \text{ rad/s}$ was chosen for simplicity).

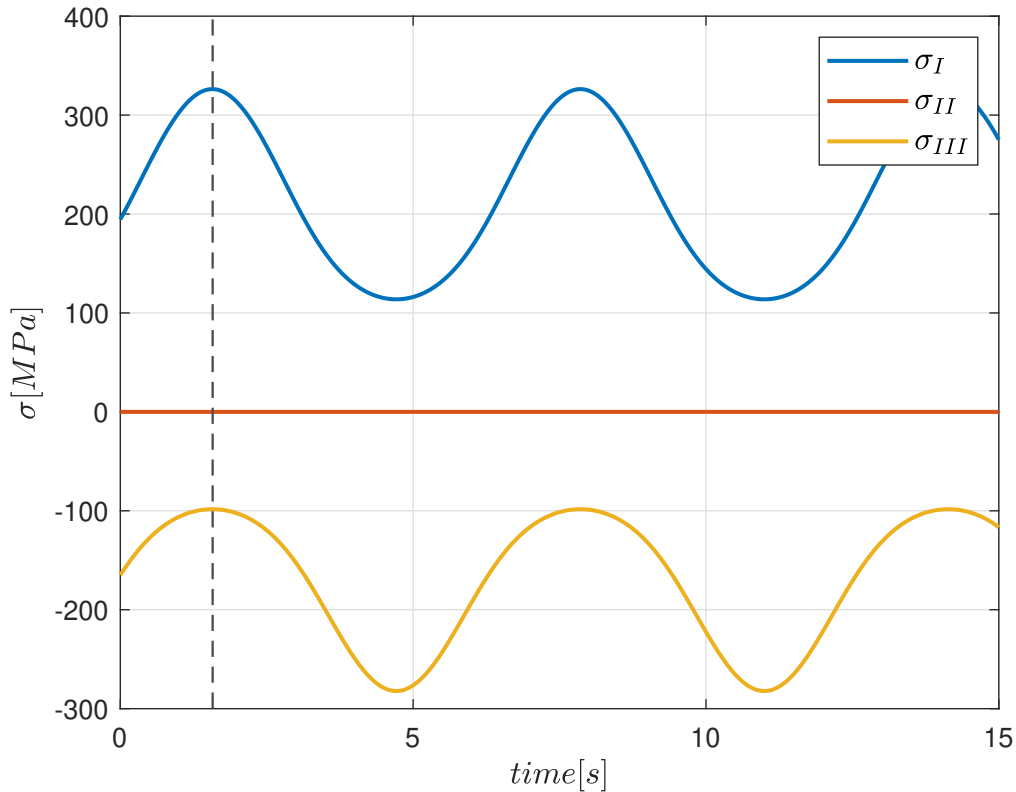


Figure 4.2: specimen fatigue assessment, principal stress components variation with time ($@\omega = 1 \text{ rad/s}$)

4.3 Static instability under compression

Since the amplitude of the alternate axial load is bigger with respect to the mean tensile load, the specimen can be subjected also to compression during fatigue test. Though it's important to verify that the specimen

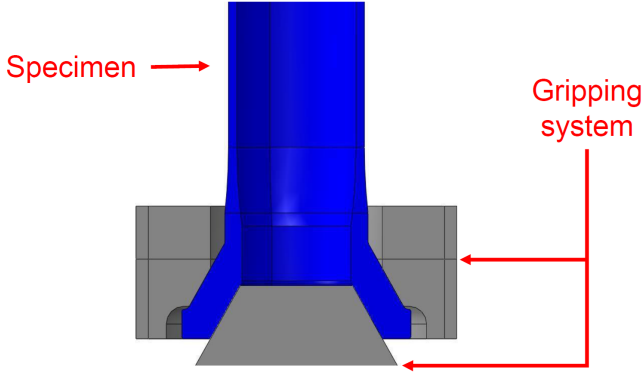


Figure 4.3: Test setup

Buckled shape of column shown by dashed line						
K	0.5	0.7	1.0	1.0	2.0	2.0

Figure 4.4: Parameter K for column effective length, American Institute of Steel Construction (1989)

doesn't reach the instability condition during those compression phases. The Euler's critical load for instability under compression can be expressed as:

$$P_{cr} = \frac{\pi^2 EI}{(KL)^2} \quad (4.6)$$

where I is the bending moment of inertia:

$$I = \frac{\pi}{64}(D^4 - d^4) = 98243 \text{ mm}^4 \quad (4.7)$$

L is the length of the column (E the Young's modulus) and K is a parameter accounting for the type of boundary conditions constraining the specimen.

Looking at the setup in figure 4.3, the length to be considered in this case is the total length of the specimen minus the length corresponding to the two conical portions bounded by the gripping devices, considered rigid (and thus preventing those parts from playing a role in the stability evaluation):

$$L = l - 2 \times l_{grip} = 460 \text{ mm} - 2 \times 40 \text{ mm} = 380 \text{ mm} \quad (4.8)$$

K must be selected according to the scheme in figure 4.4: for the specific case, the gripping devices must be considered as clamps, because they prevent any rotation of the section at the extremities and thus $K = 0.5$.

We obtain a critical load equal to:

$$P_{cr} = 5371.8 \text{ kN} \quad (4.9)$$

While the maximum compressive load during fatigue test is:

$$N_{min} = N_m - \max(N_a \sin(\omega t)) = -68 \text{ kN} \quad (4.10)$$

For avoiding the onset of a possible instability, the following condition must be fulfilled:

$$|N_{min}| << \frac{P_{cr}}{\eta} \quad (4.11)$$

obtaining the following safety factor for critical load instability:

$$\eta_{cr} = \frac{P_{cr}}{|N_{min}|} = 79.0 \quad (4.12)$$

As expected, the safety factor is very big, since for this kind of test it's necessary to stress the specimen with loads which are very far from potentially giving rise to instability phenomena, in order for the fatigue test to be meaningful.

4.4 Fatigue notch coefficient

Since a fatigue loading condition is present, it is mandatory to take in consideration, in doing the assessment, some coefficient of surface finish m_s , the coefficient of dimension m_d and the fatigue notch factor K_f .

Starting from the surface finish effect we the following data can be used:

- σ_{UTS} equal to $1020MPa$ for 9310 VIMVAR and $1000MPa$ 32CDV13 material
- $R_a = 0,4\mu m$

Where the roughness is identified by considering that the specimen has been subjected to grinding and using the graph in figure 4.5, a mean value of $0,4\mu m$ is retrieved.

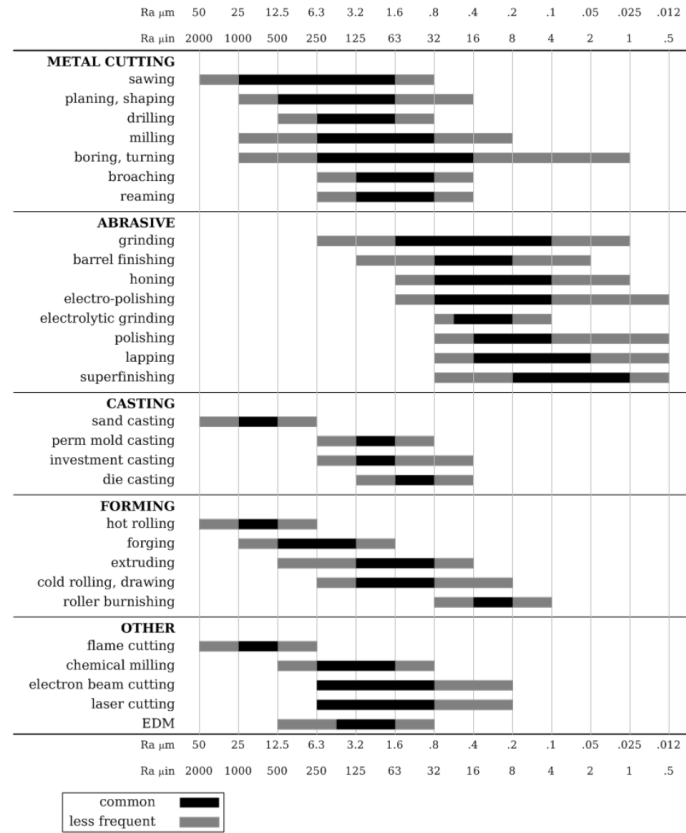


Figure 4.5: Roughness given by different surface finishing procedure

So with the diagram of the m_s a value of $m_s = 1$ can be assumed for both material.

Then for the dimension effect we can directly consider $m_d = 1$ for both material since the specimen loading condition involve only axial fatigue.

And lastly, regarding the K_f we can use the formula:

$$K_{ff} = 1 + q (K_{tf} - 1)$$

$$K_{ft} = 1 + q (K_{tt} - 1)$$

But since has been considered, as in the static assessment of the specimen, K_{tf} and K_{tt} equal to 1, it means that K_{ff} and K_{ft} are unitary again for both material.

4.5 Verification of Resistance: Sines's Criterion

In order to complete the fatigue assessment, the Sines criterion is employed. This criterion is based on the following inequality:

$$\frac{\sigma_s^*}{1 - \frac{I_{1,m}}{\sigma_{UTS}}} \leq \frac{\sigma_{\lim}}{\eta}$$

where the quantities involved are defined as follows:

- σ_s^* is the equivalent fully reversed alternating stress, computed using the Von Mises criterion for the alternating components:

$$\sigma_s^* = \sigma_{VM,a} = \sqrt{\sigma_{Ia}^2 + \sigma_{IIa}^2 + \sigma_{IIIa}^2 - \sigma_{Ia}\sigma_{IIa} - \sigma_{IIa}\sigma_{IIIa} - \sigma_{IIIa}\sigma_{Ia}} \approx \sqrt{\sigma_a^2 + 3\tau_a^2} = 198.11 \text{ MPa}$$

- $I_{1,m}$ is the first invariant of the mean stress tensor, i.e., the hydrostatic part of the mean stress:

$$I_{1,m} = \sigma_{Im} + \sigma_{IIIm} + \sigma_{IIIIm} = \sigma_m = 29.72 \text{ MPa}$$

- $\sigma_{\lim}$ is the fatigue limit corrected by technological and geometrical factors:

$$\sigma_{\lim} = \sigma_{FA,a} \cdot m_s \cdot m_d \cdot \frac{1}{K_{ff}} = \sigma_{FA,a}$$

where $\sigma_{FA,a} = 350 \text{ MPa}$ for 9310 VIMVAR and $\sigma_{FA,a} = 320 \text{ MPa}$ for 32CDV13 material.

Therefore, the calculated safety factors η for the two materials are:

- $\eta = 1.715$ for 9310 VIMVAR
- $\eta = 1.567$ for 32CDV13

5 Comparison with FEM and experiments

5.1 Acquisition of Data from Strain Gauge Rosette

Strain gauges are precision sensors used to measure the deformation (strain) of mechanical components under applied loads. They operate by converting mechanical strain into a measurable change in electrical resistance, which can then be correlated to the strain experienced by the material. Strain gauges are extensively employed in structural health monitoring, material characterization, and mechanical design to assess stress distributions and predict potential failure.

In this study, strain gauges were used to evaluate the stress state of our specimen, a standard procedure in the field of Machine Design. Two set of three-element strain gauge (SG) rosette was mounted on the specimen to capture multi-axial strain data during loading. The data from three independent experimental campaigns were recorded in three separate “.txt” files, each organized into eleven columns, with each column representing a specific physical quantity recorded over discrete time steps.

Due to the large volume of data collected, it was necessary to import the datasets into a spreadsheet environment to facilitate structured and efficient analysis. A MATLAB script was developed to extract and process the experimental stress data from the SG rosette measurements. This script enabled the calculation of stress components over time for each of the three tests performed on the same specimen.

Figure ?? illustrates the time evolution of the key physical quantities measured during the three experimental tests (denoted as A, B, and C). The plots display axial displacement, axial force, torque angle, torque moment, and six auxiliary input channels, which correspond to the axial strain for each SG. These time series provide a comprehensive overview of the loading conditions and response behavior of the specimen throughout each test campaign (at every time step).

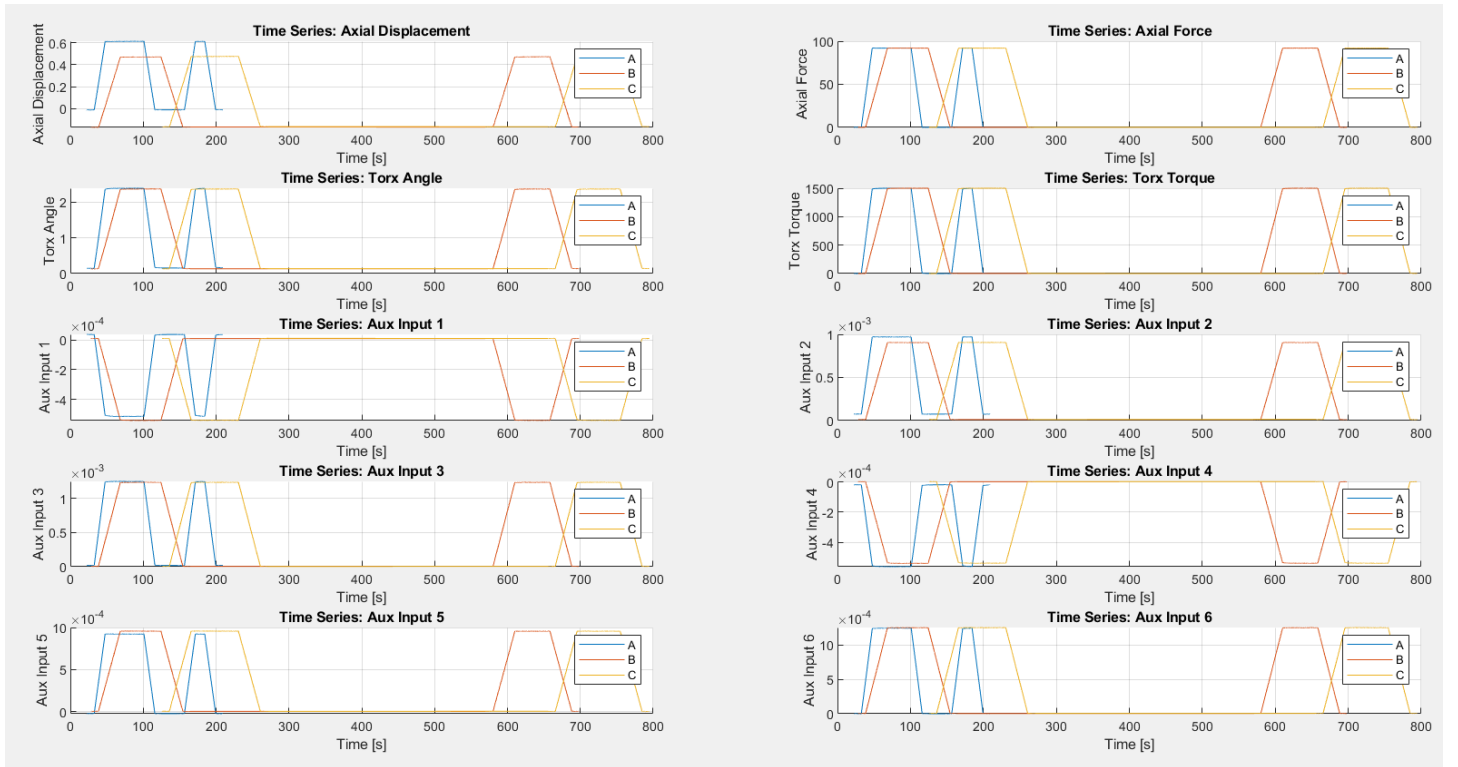


Figure 5.1: comparison between the time evolution of each physical quantity as function of time

The following section presents the time-dependent results (5.2, 5.3, 5.4) from the three experimental tests, highlighting now only:

- axial displacement;

- axial load;
- torque angle;
- torque moment;

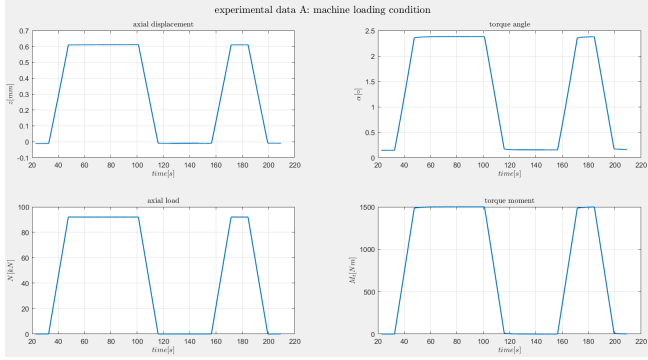


Figure 5.2: Experimental data A

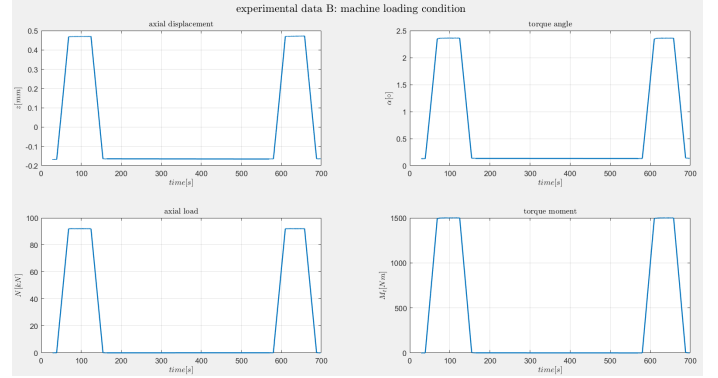


Figure 5.3: Experimental data B

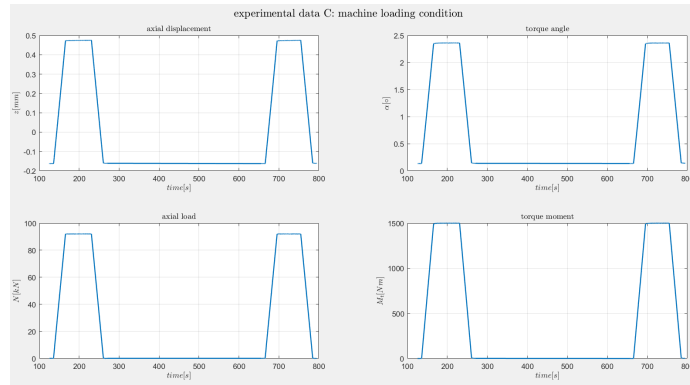


Figure 5.4: Experimental data C

The goal of this analysis is to evaluate the state of strain at a specific critical time, and subsequently compute the state of stress and Von Mises equivalent stress. This approach allows the transformation of strain measurements into meaningful stress information, which is then used to validate analytical and numerical models.

The input data consisted of time histories of the following quantities:

- Axial displacement (z) in mm;
- Axial force (N) in kN;
- Torque angle (α) in degrees;
- Torque moment (M_t) in Nm;
- Strains recorded by three strain gauges positioned at $+45^\circ$, 0° and -45° .

To analyze the critical load condition, the script identifies the time instant corresponding to the **maximum axial force** by detecting peaks in the load signal. This ensures that the computed state of stress corresponds to the most demanding condition experienced by the specimen during the test.

Strain Rosette Analysis. A standard three-element strain gauge rosette is used to measure in-plane (2D) strains. The measured strains (ε_a , ε_b , ε_c) correspond to directions at $+45^\circ$, 0° , and -45° , respectively and are the input to reconstruct the entire state of stress. Indeed these values are used to solve the strain transformation equations. By adopting the following rotation matrix A :

$$A = \begin{bmatrix} \cos^2 \theta_a & \sin^2 \theta_a & \sin \theta_a \cos \theta_a \\ \cos^2 \theta_b & \sin^2 \theta_b & \sin \theta_b \cos \theta_b \\ \cos^2 \theta_c & \sin^2 \theta_c & \sin \theta_c \cos \theta_c \end{bmatrix}$$

The linear system can be written as:

$$A \begin{bmatrix} \varepsilon_z \\ \varepsilon_x \\ \gamma_{zx} \end{bmatrix} = \begin{bmatrix} \varepsilon_a \\ \varepsilon_b \\ \varepsilon_c \end{bmatrix}$$

Finally, solving the system yields the following result.

$$\boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_z \\ \varepsilon_x \\ \gamma_{zx} \end{bmatrix} = A^{-1} \begin{bmatrix} \varepsilon_a \\ \varepsilon_b \\ \varepsilon_c \end{bmatrix}$$

In this way, we are now able to obtain:

- ε_x : strain along the axial direction of the specimen;
- ε_z : strain in the radial direction of the specimen;
- γ_{zx} : in-plane shear strain of the specimen.

The strain along y direction can be computed starting from the ones in the other two directions and exploiting the knowledge that we are working with a plane state of stress. To fulfill this condition, the following strain is needed:

$$\varepsilon_y = \frac{-\nu}{1-\nu}(\varepsilon_x + \varepsilon_z) \quad (5.1)$$

Constitutive Model. Assuming the material behaves as an isotropic, linear elastic solid, the full stress state is calculated using Hooke's law in three dimensions. The constitutive relation is defined as:

$$\boldsymbol{\sigma} = \mathbf{C} \cdot \boldsymbol{\varepsilon}$$

where $\mathbf{C}$ is the 6x6 stiffness matrix constructed from Young's modulus E and Poisson's ratio ν of the selected material. In detail, the elasticity matrix $\mathbf{C}$ in Voigt notation is given by:

$$\mathbf{C} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & \nu & 0 & 0 & 0 \\ \nu & 1-\nu & \nu & 0 & 0 & 0 \\ \nu & \nu & 1-\nu & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1-2\nu}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1-2\nu}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1-2\nu}{2} \end{bmatrix}$$

From the computed stress components, the full 3D stress tensor is reconstructed:

$$\boldsymbol{\sigma} = \begin{bmatrix} \sigma_x & 0 & \tau_{zx} \\ 0 & \sigma_y & 0 \\ \tau_{zx} & 0 & \sigma_z \end{bmatrix}$$

The eigenvalues of this tensor provide the three principal stresses σ_I , σ_{II} , and σ_{III} . These represent the normal stresses acting on planes where shear components vanish.

Finally, to evaluate the yielding condition, the Von Mises stress is computed as:

$$\sigma_{VM} = \sqrt{\sigma_I^2 + \sigma_{II}^2 + \sigma_{III}^2 - \sigma_I\sigma_{II} - \sigma_{II}\sigma_{III} - \sigma_{III}\sigma_I}$$

5.2 Statistical analysis of the data

To efficiently handle the extensive volume of experimental data and extract meaningful statistical insights, a dedicated MATLAB script was developed. This routine automates the import of data from the three testing campaigns and performs a comparative statistical analysis for each measured physical parameter. For every quantity, the script generates a scatter plot, highlighting the distribution of raw data points across the three tests. Moreover, we adopted the well known Gaussian probability density function (PDF) to visualize the underlying statistical distribution. This approach provides both a qualitative and quantitative understanding of the data variability and consistency across campaigns A, B, and C. This type of statistical evaluation is widely adopted in the aerospace industry, where reliability and safety are of paramount importance. By applying such methods, it becomes possible to gain a clearer understanding of the overall load and stress spectra that a component may experience throughout its service life. For illustrative purposes, a selection of representative results is presented below (5.5, 5.6, 5.7, 5.8).

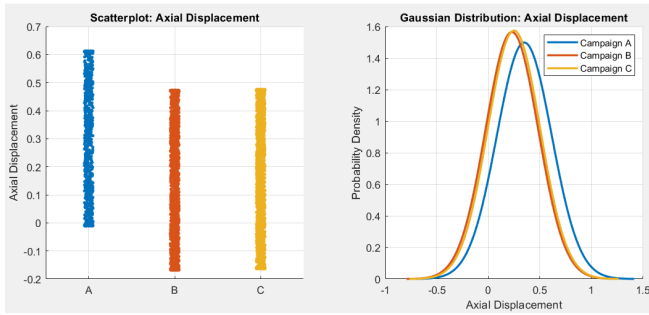


Figure 5.5: Axial displacement

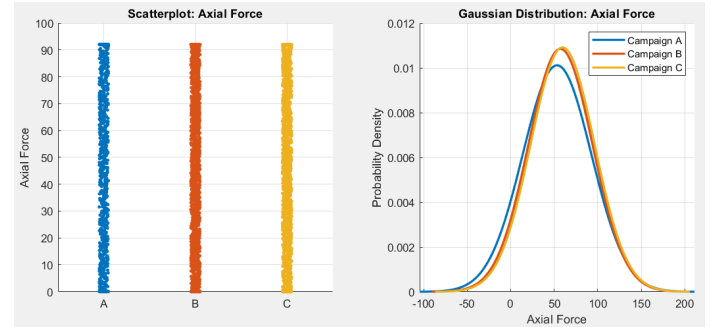


Figure 5.6: Axial force

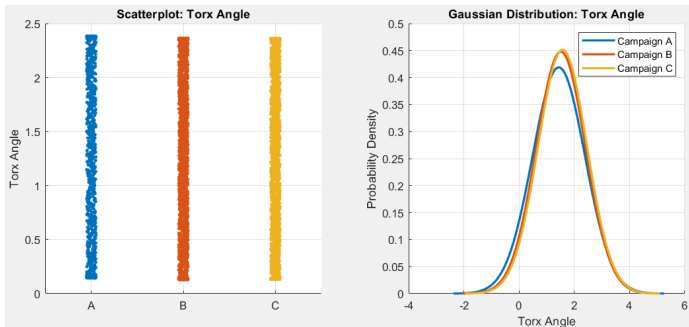


Figure 5.7: Torx angle

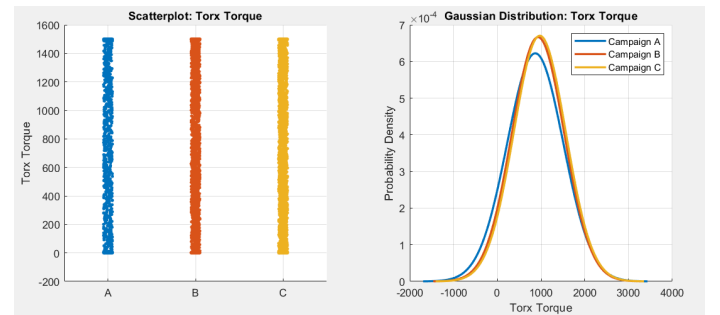


Figure 5.8: Torx torque

5.3 Analytical Results vs Experimental Results

In the previous section, we described the procedure used to compute the equivalent stress from the extensive dataset obtained during the experimental campaign. We now compare these experimental results with the analytical predictions discussed earlier in the presentation, in order to assess their level of agreement. Below are the *Von Mises* stress values obtained from each experimental campaign:

- $\sigma_{VM_{SG_A}} = 300.0348 \text{ MPa}$;
- $\sigma_{VM_{SG_B}} = 293.4037 \text{ MPa}$;

- $\sigma_{VM_{SG_C}} = 293.2517 \text{ MPa}$.

The variation among these three values is likely influenced by the positioning of the measuring instruments on the shaft, which probably differ one from each other.

As previously discussed, the following loads were applied at the free end of the specimen:

- $N(t) = N_m + N_a \sin(\omega t) = 92000 \text{ N}$: axial force as a function of time;
- $M_t = \max(M_{t,m}) = -1500 \text{ N} \cdot \text{m}$: maximum applied torque;

This represents the maximum loads that the structure may experience during fatigue testing. The resulting equivalent stress in the most stressed section is: $\sigma_{VM_{analytical}} = 384.9870 \text{ MPa}$.

As observed, the numerical result differs from the experimental values by a range between 84.9522 MPa and 91.7353 MPa . This discrepancy is likely due to the strain gauges being positioned at a different location along the specimen, possibly away from the notch.

5.4 Analytical results vs FEM results

When dealing with the design of a mechanical component, FEM (*Finite Element Method*) simulations are fundamental to predict the behavior of the structure and quantify how much analytical and mathematical models are suitable to describe the phenomenon. In this case, we have used *Abaqus*, which is a finite element analysis (FEA) software developed by *Dassault Systèmes*, used to simulate the physical behavior of structures and solid bodies under loads, temperature variations, contact, impact, and other environmental conditions. For the setting up of the simulation, we first uploaded the 3D model in the software. The shaft has been partitioned and proper values of E and ν has been chosen to describe the real material of the component. Lately it was necessary to apply constraints at one end, while on the other we put the following loads:

- $N(t) = N_m + N_a \sin(\omega t) = 92000 \text{ N}$: axial force as a function of time;
- $M_t = \max(M_{t,m}) = -1500 \text{ N} \cdot \text{m}$: maximum applied torque.

After generating a proper mesh (taking into account that this software can do calculation with a maximum number of nodes) we just launch the simulation. Within *Abaqus* it is possible to check different parameters. We focused on the Von Mises results and understand how much we get close between the FEM simulation and analytical calculations. The graphical visualization and the Von Mises curve along the longitudinal axis of the sample, obtained from the FEM analysis on Abaqus, are shown, respectively, in figures 5.9 and 5.10.

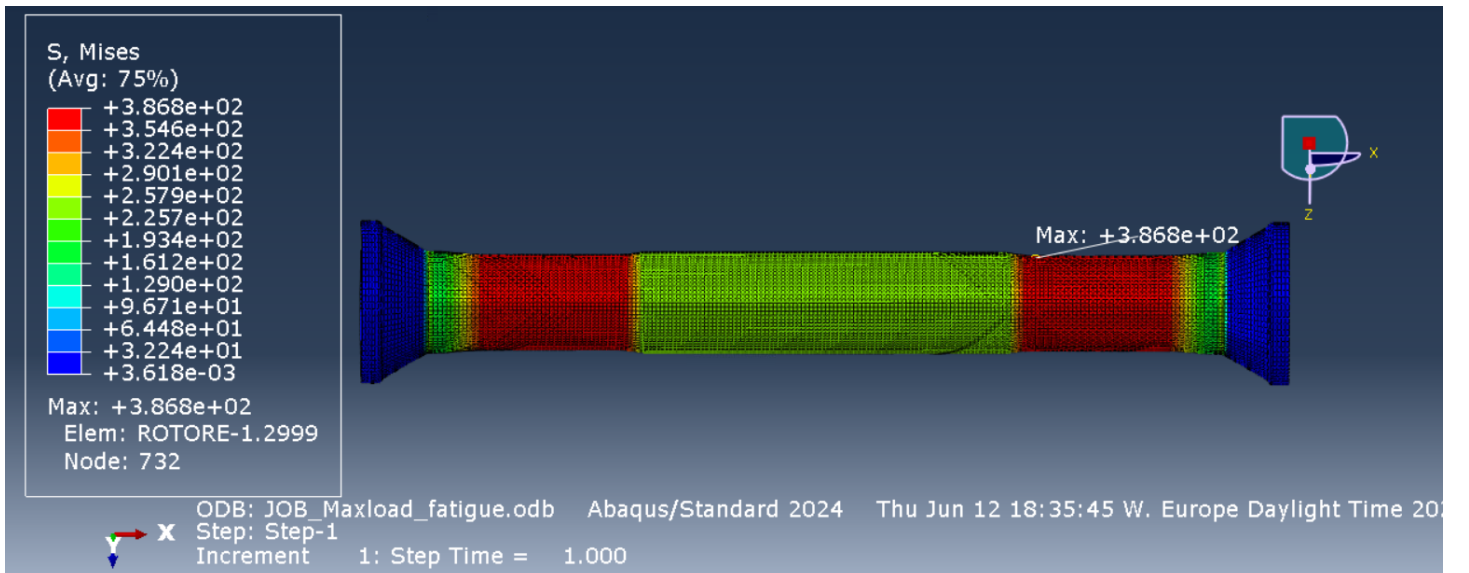


Figure 5.9: Von Mises visualization (@ max fatigue loads)

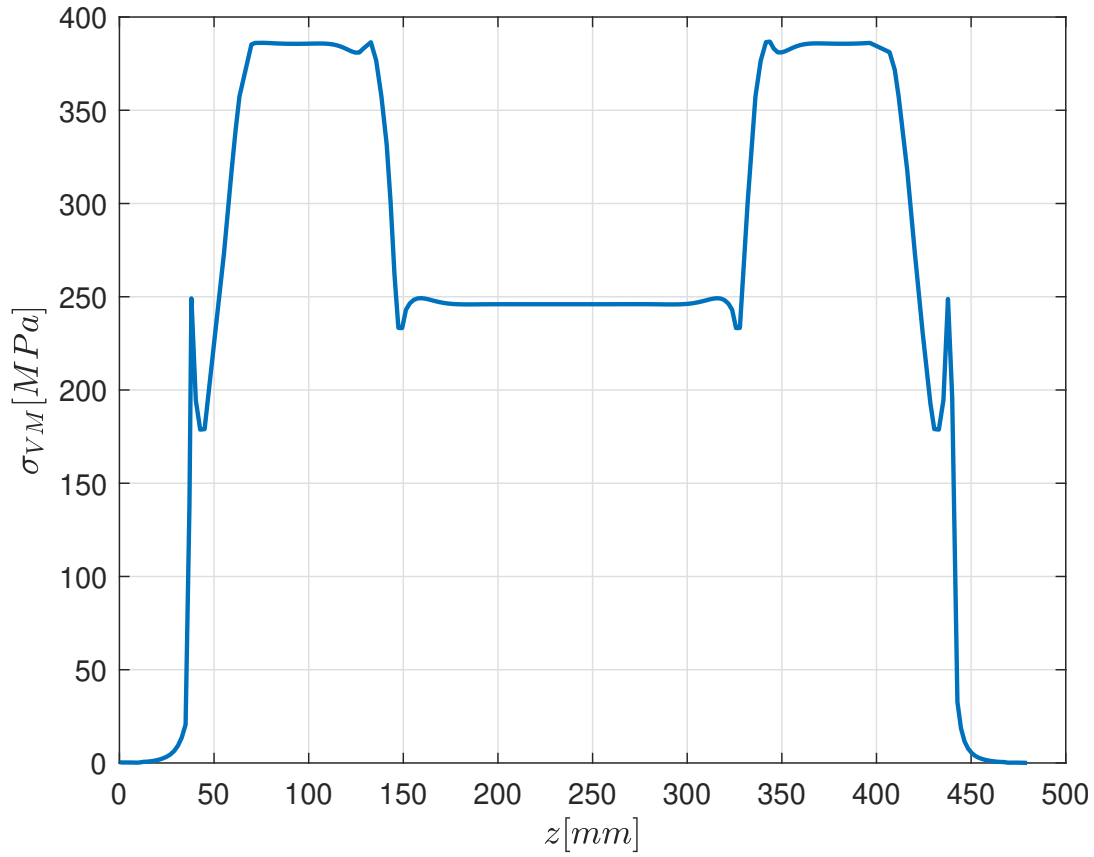


Figure 5.10: FEM Von Mises (@ max fatigue loads)

The FEM results appear to be highly reliable, as the equivalent Von Mises stress is higher in the regions of the shaft where the diameter is shorter, and decreases in the central section, where the geometry is thicker. This behavior aligns well with theoretical expectations.

In the notch region, the FEM simulation yields the following stress value:

$$\sigma_{VM,FEM} = 386.8 \text{ MPa}$$

This result is in excellent agreement with the analytical solution. The absolute difference between the two values is:

$$\Delta\sigma_{VM} = \sigma_{VM,FEM} - \sigma_{VM,analytical} = 1.813 \text{ MPa}$$

To quantify the discrepancy, the relative percentage error is computed as:

$$\delta = \left(\frac{|\sigma_{VM,FEM} - \sigma_{VM,analytical}|}{\sigma_{VM,analytical}} \right) \times 100\% = 0.47\%$$

This small percentage error confirms the high level of accuracy and consistency between the FEM simulation and the analytical approach.

6 Final conclusions

The combined numerical and experimental analysis conducted in this study allow for significant conclusions.

Firstly, both the static and fatigue assessments have been carried out on the simplified mast and on the test specimen, and considering the two materials under investigation confirmed that the adopted design criteria effectively satisfied the required structural performance targets. Analytical and finite element results has given close agreement.

Also regarding the value of maximum Von Mises stress values there is consistency, thereby validating the modeling approach and supporting the use of these simulations as a reliable basis for preliminary fatigue evaluations.

As a last consideration, although the position of the rosette sensors was not provided, it can be retrieved by comparing the equivalent Von Mises stress obtained experimentally with the one obtained with the numerical outcomes of the FEM model, meaning that the strain gauges are placed in proximity of the notches areas.

A MATLAB code

```
%=====
%                               MAST STATIC ASSESSMENT
%=====

close all
clear all
clc

%% Data
% Geometry:
R_i = 70;           % [mm] internal radius (critical section)
t = 9;              % [mm] shaft thickness (critical section)
D_e = 2*(R_i + t); % [mm] external diameter
D_i = 2*R_i;        % [mm] internal diameter
l1 = 438;           % [mm] distance from B.C.1
l2 = 202;           % [mm] distance from load application point
k_t_a = 1.12;       % [-] notch coefficient under tension
k_t_t = 1;          % [-] notch coefficient under torsion

% Materials:
matNames = ["9310 VIM VAR" "32CDV13"]; % Materials names
E = [200000 200000]; % [MPa] Young modulus
G = [76000 76000]; % [MPa] Shear modulus
nu = [0.32 0.3]; % [-] Poisson ratio
S_ys = [893 850]; % [MPa] Yielding strength
UTS = [1020 1000]; % [MPa] Ultimate tensile strength
sigma_FA_a = [350 320]; % [MPa] Endurance limit

%% Loads
N = 1e5; % [N]
T = 3.2e4; % [N]
M_b = 3.5e7; % [Nmm]
M_t = -8e7; % [Nmm]

%% Reaction forces
V1 = -M_b/l1 - T*l2/l1;
H1 = -N;
V2 = M_b/l1 + T*(1 + l2/l1);
C1 = M_t;

%% Internal forces
% Part 1
z1 = linspace(0, l1, 1e4);
N1 = -H1*ones(1, length(z1));
T1 = V1*ones(1, length(z1));
M_b1 = -V1*z1;
M_t1 = C1*ones(1, length(z1));

% Part 2
z2 = linspace(0, l2, 1e4);
N2 = -H1*ones(1, length(z2));
T2 = (V1 + V2)*ones(1, length(z2));
M_b2 = -V1*(l1 + z2) - V2*z2;
M_t2 = C1*ones(1, length(z2));

%% Internal forces diagrams
```

```

col = rgb2hex(orderedcolors("gem"));
col = col(1); % diagrams color
opacity = 0.80; % diagrams opacity
FS = 12; % Font Size

z_beam = linspace(0, l1+l2, 1e4); % beam length vector

fig1 = figure('Position', get(0, 'Screensize'));
% sgtitle('$MAST \ static \ assessment: \ internal \ forces \ diagrams$', '
    Interpreter', 'latex', 'FontSize', FS);

subplot(2,2,1);
area(z1, N1, 'FaceColor', col, 'FaceAlpha', opacity);
hold on;
grid on;
area(z2+l1, N2, 'FaceColor', col, 'FaceAlpha', opacity);
plot(z_beam, zeros(1, length(z_beam)), 'k', 'LineWidth', 2);
ylim_value = max(max(abs(N1)), max(abs(N2)));
ylim([-2*ylim_value 2*ylim_value]);
xlabel('$z$ [mm]$', 'Interpreter', 'latex', 'FontSize', FS);
ylabel('$N$ [N]$', 'Interpreter', 'latex', 'FontSize', FS);
title('$Axial \ force$', 'Interpreter', 'latex', 'FontSize', FS);

subplot(2,2,2);
area(z1, T1, 'FaceColor', col, 'FaceAlpha', opacity);
hold on;
grid on;
area(z2+l1, T2, 'FaceColor', col, 'FaceAlpha', opacity);
plot(z_beam, zeros(1, length(z_beam)), 'k', 'LineWidth', 2);
ylim_value = max(max(abs(T1)), max(abs(T2)));
ylim([-2*ylim_value 2*ylim_value]);
xlabel('$z$ [mm]$', 'Interpreter', 'latex', 'FontSize', FS);
ylabel('$T$ [N]$', 'Interpreter', 'latex', 'FontSize', FS);
title('$Shear \ force$', 'Interpreter', 'latex', 'FontSize', FS);

subplot(2,2,3);
area(z1, M_b1, 'FaceColor', col, 'FaceAlpha', opacity);
hold on;
grid on;
area(z2+l1, M_b2, 'FaceColor', col, 'FaceAlpha', opacity);
plot(z_beam, zeros(1, length(z_beam)), 'k', 'LineWidth', 2);
ylim_value = max(max(abs(M_b1)), max(abs(M_b2)));
ylim([-2*ylim_value 2*ylim_value]);
xlabel('$z$ [mm]$', 'Interpreter', 'latex', 'FontSize', FS);
ylabel('$M_f$ [N]$', 'Interpreter', 'latex', 'FontSize', FS);
title('$Bending \ moment$', 'Interpreter', 'latex', 'FontSize', FS);

subplot(2,2,4);
area(z1, M_t1, 'FaceColor', col, 'FaceAlpha', opacity);
hold on;
grid on;
area(z2+l1, M_t2, 'FaceColor', col, 'FaceAlpha', opacity);
plot(z_beam, zeros(1, length(z_beam)), 'k', 'LineWidth', 2);
ylim_value = max(max(abs(M_t1)), max(abs(M_t2)));
ylim([-2*ylim_value 2*ylim_value]);
xlabel('$z$ [mm]$', 'Interpreter', 'latex', 'FontSize', FS);
ylabel('$M_t$ [N]$', 'Interpreter', 'latex', 'FontSize', FS);
title('$Torque \ moment$', 'Interpreter', 'latex', 'FontSize', FS);

exportgraphics(fig1, 'outputFig\MAST_internal_forces_diagrams.eps', 'ContentType', '
    vector');

```

```

%% Critical section
N_c = N; % axial force at the critical section
M_t_c = M_t; % torque moment at the critical section
M_b_c = M_b1(end); % bending moment at the critical section

%% Static assessment
sigma = (N_c*4/(pi*(D_e^2 - D_i^2)) + (32*M_b_c*D_e)/(pi*(D_e^4 - D_i^4)));
sigma_notch = sigma*k_t_a;
tau = (16 * abs(M_t_c) * D_e) / ((pi*(D_e^4 - D_i^4))); % tau_notch = tau -> k_t_t = 1

stress_tensor = [0 0 tau; 0 0 0; tau 0 sigma_notch];
principal_stresses = sort(eig(stress_tensor), 'descend');
sigma_I = principal_stresses(1);
sigma_II = principal_stresses(2);
sigma_III = principal_stresses(3);

% Guest-Tresca
sigma_GT = sigma_I - sigma_III;

% Von Mises
sigma_VM = sqrt(sigma_I^2 + sigma_II^2 + sigma_III^2 - sigma_I*sigma_II - sigma_II*sigma_III - sigma_I*sigma_III);
% sigma_VM = sqrt(sigma_notch^2 + 3*tau^2); % simplified Von Mises

eta = S_ys ./ [sigma_GT sigma_VM]'; % safety factor

T = table(eta(:,1), eta(:,2), 'RowNames', ["Guest-Tresca" "Von Mises"], 'VariableNames', matNames);
disp(' (safety factor) - MAST static assessment:');
disp(T);

%=====
% SPECIMEN STATIC ASSESSMENT
%=====

close all
clear all
clc

%% Data
% Geometry:
D_e = 46.94; % [mm] external diameter
D_i = 41.1; % [mm] internal diameter
l = 460; % [mm] length of the specimen
k_t_a = 1; % [-] notch coefficient under tension
k_t_t = 1; % [-] notch coefficient under torsion

% Materials:
matNames = ["9310 VIM VAR" "32CDV13"]; % Materials names
E = [200000 200000]; % [MPa] Young modulus
G = [76000 76000]; % [MPa] Shear modulus
nu = [0.32 0.3]; % [-] Poisson ratio
S_ys = [893 850]; % [MPa] Yielding strength
UTS = [1020 1000]; % [MPa] Ultimate tensile strength
sigma_FA_a = [350 320]; % [MPa] Endurance limit

%% Loads
N = 2.9e4; % [N]
M_b = 1.019e6; % [Nmm]
M_t = -2.328e6; % [Nmm]

```

```

%% Internal forces
z = linspace(0, 1, 1e4);
N_int = N*ones(1, length(z));
T_int = 0*ones(1, length(z));
M_b_int = M_b*ones(1, length(z));
M_t_int = M_t*ones(1, length(z));

%% Internal forces diagrams
col = rgb2hex(orderedcolors("gem"));
col = col(1); % diagrams color
opacity = 0.9; % diagrams opacity

z_beam = linspace(0, 1, 1e4); % beam length vector

fig1 = figure('Position', get(0, 'Screensize'));
% sgtitle('$Specimen \ static \ assessment: \ internal \ forces \ diagrams$', '
    Interpreter', 'latex', 'FontSize', FS);
FS = 12; % Font Size

subplot(2,2,1);
area(z, N_int, 'FaceColor', col, 'FaceAlpha', opacity);
hold on;
grid on;
plot(z_beam, zeros(1, length(z_beam)), 'k', 'LineWidth', 2);
ylim_value = max(abs(N_int));
ylim([-2*max(1, ylim_value), 2*max(1, ylim_value)]);
xlabel('$z$ [mm]$', 'Interpreter', 'latex', 'FontSize', FS);
ylabel('$N$ [N]$', 'Interpreter', 'latex', 'FontSize', FS);
title('$Axial \ force$', 'Interpreter', 'latex', 'FontSize', FS);

subplot(2,2,2);
area(z, T_int, 'FaceColor', col, 'FaceAlpha', opacity);
hold on;
grid on;
plot(z_beam, zeros(1, length(z_beam)), 'k', 'LineWidth', 2);
ylim_value = max(abs(T_int));
ylim([-2*max(1, ylim_value), 2*max(1, ylim_value)]);
xlabel('$z$ [mm]$', 'Interpreter', 'latex', 'FontSize', FS);
ylabel('$T$ [N]$', 'Interpreter', 'latex', 'FontSize', FS);
title('$Shear \ force$', 'Interpreter', 'latex', 'FontSize', FS);

subplot(2,2,3);
area(z, M_b_int, 'FaceColor', col, 'FaceAlpha', opacity);
hold on;
grid on;
plot(z_beam, zeros(1, length(z_beam)), 'k', 'LineWidth', 2);
ylim_value = max(abs(M_b_int));
ylim([-2*max(1, ylim_value), 2*max(1, ylim_value)]);
xlabel('$z$ [mm]$', 'Interpreter', 'latex', 'FontSize', FS);
ylabel('$M_f$ [N]$', 'Interpreter', 'latex', 'FontSize', FS);
title('$Bending \ moment$', 'Interpreter', 'latex', 'FontSize', FS);

subplot(2,2,4);
area(z, M_t_int, 'FaceColor', col, 'FaceAlpha', opacity);
hold on;
grid on;
plot(z_beam, zeros(1, length(z_beam)), 'k', 'LineWidth', 2);
ylim_value = max(abs(M_t_int));
ylim([-2*max(1, ylim_value), 2*max(1, ylim_value)]);
xlabel('$z$ [mm]$', 'Interpreter', 'latex', 'FontSize', FS);
ylabel('$M_t$ [N]$', 'Interpreter', 'latex', 'FontSize', FS);

```

```

title('$Torque \ moment$', 'Interpreter', 'latex', 'FontSize', FS);

exportgraphics(fig1, 'outputFig\Specimen_internal_forces_diagrams.eps', 'ContentType',
    'vector');

%% Critical section
N_c = N;          % axial force at the critical section
M_t_c = M_t;      % torque moment at the critical section
M_b_c = M_b;      % bending moment at the critical section

%% Static assessment
sigma = N_c*4/(pi*(D_e^2 - D_i^2)) + (32*M_b_c*D_e)/(pi*(D_e^4 - D_i^4));
sigma_notch = sigma*k_t_a; % k_t_a = 1
tau = (16 * abs(M_t_c) * D_e) / ((pi*(D_e^4 - D_i^4)));
tau_notch = tau*k_t_t; % k_t_t = 1

stress_tensor = [0 0 tau_notch; 0 0 0; tau_notch 0 sigma_notch];
principal_stresses = sort(eig(stress_tensor), 'descend');
sigma_I = principal_stresses(1);
sigma_II = principal_stresses(2);
sigma_III = principal_stresses(3);

% Guest-Tresca
sigma_GT = sigma_I - sigma_III;

% Von Mises
sigma_VM = sqrt(sigma_I^2 + sigma_II^2 + sigma_III^2 - sigma_I*sigma_II - sigma_II*
    sigma_III - sigma_I*sigma_III);
% sigma_VM = sqrt(sigma_notch^2 + 3*tau^2); % simplified Von Mises

eta = S_ys ./ [sigma_GT sigma_VM]'; % safety factor

T = table(eta(:,1), eta(:,2), 'RowNames', ["Guest-Tresca" "Von Mises"], '
    VariableNames', matNames);
disp("    (safety factor) - specimen static assessment:");
disp(T);

%% Equivalent axial load
% Equivalent stress generated by bending moment, which the machine cannot
% apply to the specimen
N_eq = 8*M_b*D_e*(D_e^2 - D_i^2)/(D_e^4 - D_i^4);
N_tot_eq = N + N_eq;

%=====
%                               SPECIMEN FATIGUE ASSESSMENT
%=====

close all
clear all
clc

%% Data
% Geometry:
D = 46.94;          % [mm] external diameter
d = 41.10;          % [mm] internal diameter
l = 460;            % [mm] specimen total length
l_grip = 40;        % [mm] length of each gripping area
l_unsup = l - 2*l_grip; % [mm] unsupported length of specimen
I = pi/64*(D^4 - d^4); % [mm^4] bending moment of inertia

% Materials:

```

```

matNames = ["9310 VIM VAR" "32CDV13"]; % Materials names
E = [200000 200000]; % [MPa] Young modulus
G = [76000 76000]; % [MPa] Shear modulus
nu = [0.32 0.3]; % [-] Poisson ratio
S_ys = [893 850]; % [MPa] Yielding strength
UTS = [1020 1000]; % [MPa] Ultimate tensile strength
sigma_FA_a = [350 320]; % [MPa] Endurance limit

%% Loads
% N = N_m + N_a*sin(w*t)
N_m = 12000; % [N] mean axial load
N_a = 80000; % [N] alternate axial load
N_min = N_m - N_a; % [N] maximum static compression load faced during fatigue test
M_t = 1.5e6; % [Nmm] torque load
w = 1; % [rad/s] alternate load pulsation

%% Instability assessment
% Eulerian instability load
l_crit = 0.5*l_unsup; % [mm] effective length, considering BCs (clamp at both
ends)
P_cr = pi^2*E(1)*I/l_crit^2; % [N] critical load (same E for both materials)
eta_crit = P_cr/abs(N_min); % [-] safety factor for Eulerian instability
disp(['Specimen instability assessment (safety factor): _crit = ' num2str(eta_crit)
]);

%% Fatigue assessment
sigma_m = N_m*4/pi/(D^2 - d^2); % mean axial stress
sigma_a = N_a*4/pi/(D^2 - d^2); % alternate axial stress
tau = 16*M_t*D/pi/(D^4 - d^4); % shear stress

syms t; % time as symbolic variable
time = linspace(0, 15, 1e3); % evaluated time interval

sigma_m = [0 0 -tau; 0 0 0; -tau 0 sigma_m]; % mean stress tensor

sigma_a_temp = [0 0 0; 0 0 0; 0 0 sigma_a]; % (temporary name for) alternate
stress tensor
sigma_a_t = [0 0 0; 0 0 0; 0 0 sigma_a*sin(w*t)]; % alternate stress tensor as
function of time
sigma_a = sigma_a_temp; % renaming the temporary alternate
stress tensor

sigma_tot = sigma_m + sigma_a; % total stress tensor
sigma_tot_t = sigma_m + sigma_a_t; % total stress tensor as function
of time

% Principal stresses of total stress tensor as function of time
sigma_princ_tot_t = eig(sigma_tot_t);
sigma_I_tot_t = double(subs(sigma_princ_tot_t(3), t, time));
sigma_II_tot_t = double(subs(sigma_princ_tot_t(1), t, time));
sigma_III_tot_t = double(subs(sigma_princ_tot_t(2), t, time));

FS = 12; % al text set to the same Font Size
fig1 = figure();
% fig1 = figure('Position', get(0, 'Screensize'));

plot(time, sigma_I_tot_t, time, sigma_II_tot_t, time, sigma_III_tot_t, 'LineWidth',
1.5);
xlabel('$time [s]$', 'Interpreter', 'latex', 'FontSize', FS);
ylabel('$\sigma [MPa]$', 'Interpreter', 'latex', 'FontSize', FS);
[pks, locs] = findpeaks(sigma_I_tot_t);
t_peak = time(locs(1));

```

```

xline(t_peak, '--k', 'LineWidth', 0.85);
grid on;
% title(['SPECIMEN FATIGUE: principal stresses (@ $\omega$ = ' num2str(w) ' \, [rad/s
    ]$)'], 'Interpreter', 'latex');
legend('$\sigma_{I}$', '$\sigma_{II}$', '$\sigma_{III}$', 'Interpreter', 'latex', '
    FontSize', 12);

exportgraphics(fig1, 'outputFig\specimen_princ_stress_func_t.eps', 'ContentType', '
    vector');

% Principal stresses of mean stress tensor
sigma_princ_m = sort(eig(sigma_m), 'descend');
sigma_I_m = sigma_princ_m(1);
sigma_II_m = sigma_princ_m(2);
sigma_III_m = sigma_princ_m(3);

% Principal stresses of alternate stress tensor
sigma_princ_a = sort(eig(sigma_a), 'descend');
sigma_I_a = sigma_princ_a(1);
sigma_II_a = sigma_princ_a(2);
sigma_III_a = sigma_princ_a(3);

% SINES' criterion
% Equivalent Sines
sigma_S = sqrt(sigma_I_a^2 + sigma_II_a^2 + sigma_III_a^2 - (sigma_I_a*sigma_II_a +
    sigma_I_a*sigma_III_a + sigma_II_a*sigma_III_a));
% sigma_S = sigma_I_a; % since sigma_II_a = sigma_III_a = 0

I_1 = sigma_I_m + sigma_II_m + sigma_III_m; % first invariant

b2 = 1; % [-] dimensional coefficient
b3 = 1; % [-] surface roughness coefficient
% k_f = 1 (since k_t_a = k_t_t = 1)
sigma_FA_a_p = sigma_FA_a * b2 * b3; % endurance limit of the real specimen

eta_FA = sigma_FA_a_p./sigma_S.*(1 - I_1./UTS);

T = table(eta_FA(1), eta_FA(2), 'VariableNames', matNames);
disp("_FA (fatigue safety factor) - specimen fatigue assessment:");
disp(T);

%=====
%
%                               EXPERIMENTAL DATA
%=====

close all
clear all
clc

%% Select file
[runName, path] = uigetfile('*.txt*');
D = readtable([path runName], 'VariableNamingRule', 'Preserve');
clear("path");
runName = erase(runName, ".txt");
varNames = D.Properties.VariableNames;
D = table2array(D);

%% Data
% Materials:
matNames = ["9310 VIM VAR" "32CDV13"]; % Materials names
E = [200000 200000]; % [MPa] Young modulus

```

```

G = [76000 76000]; % [MPa] Shear modulus
nu = [0.32 0.3]; % [-] Poisson ratio
S_ys = [893 850]; % [MPa] Yielding strength
UTS = [1020 1000]; % [MPa] Ultimate tensile strength
sigma_FA_a = [350 320]; % [MPa] Endurance limit

%% Data analysis
figure();
sgtitle([runName ': machine loading condition'], 'Interpreter', 'latex');

subplot(2,2,1);
plot(D(:,1), D(:,2), 'LineWidth', 1.5);
xlabel('$time [s]$', 'Interpreter', 'latex');
ylabel('$z [mm]$', 'Interpreter', 'latex');
grid on;
title('axial displacement', 'Interpreter', 'latex');

subplot(2,2,3);
plot(D(:,1), D(:,3), 'LineWidth', 1.5);
xlabel('$time [s]$', 'Interpreter', 'latex');
ylabel('$N [kN]$', 'Interpreter', 'latex');
grid on;
title('axial load', 'Interpreter', 'latex');

subplot(2,2,2);
plot(D(:,1), D(:,4), 'LineWidth', 1.5);
xlabel('$time [s]$', 'Interpreter', 'latex');
ylabel('$\alpha [\circ]$', 'Interpreter', 'latex');
grid on;
title('torque angle', 'Interpreter', 'latex');

subplot(2,2,4);
plot(D(:,1), D(:,5), 'LineWidth', 1.5);
xlabel('$time [s]$', 'Interpreter', 'latex');
ylabel('$M_t [Nm]$', 'Interpreter', 'latex');
grid on;
title('torque moment', 'Interpreter', 'latex');

%%
l = 380; % [mm] length of the specimen w/o gripping area
ad = D(:,2); % [mm] axial displacement

[pks, locs] = findpeaks(D(:,2), 'MinPeakProminence', 0.5);
[Max_value, iMax] = max(pks);
t_peak = D(locs(iMax), 1);

time = t_peak; % [s] select time instant
toll = 1e-5; % tolerance over the chosen time instant
iRos = 1; % select rosette (1 for first rosette, 2 for second one)

time = 100; % [s] select time instant
toll = 1e-5; % tolerance over the chosen time instant
iRos = 1; % select rosette (1 for first rosette, 2 for second one)

iTime = min(find(abs(D(:,1) - time) < toll));
if isempty(iTime)
    error('Time instant not found: increase tolerance!');
end

if iRos == 1
    eps_a = D(iTime,6);
    eps_b = D(iTime,7);

```

```

        eps_c = D(iTime,8);
elseif iRos == 2
        eps_a = D(iTime,9);
        eps_b = D(iTime,10);
        eps_c = D(iTime,11);
end

theta_a = 45; % [ ] strain gauge A orientation
theta_b = 0; % [ ] strain gauge A orientation
theta_c = -45; % [ ] strain gauge A orientation

% Matrix of the linear system
A = [(cosd(theta_a))^2 (sind(theta_a))^2 (sind(theta_a))*(cosd(theta_a));
      (cosd(theta_b))^2 (sind(theta_b))^2 (sind(theta_b))*(cosd(theta_b));
      (cosd(theta_c))^2 (sind(theta_c))^2 (sind(theta_c))*(cosd(theta_c))];

% Known terms vector of the linear system
b = [eps_a; eps_b; eps_c];

% Linear system solution
x = A\b;

eps_z = x(1);
eps_x = x(2);
gamma_zx = x(3);

% Select material
iMat = 1;

E = E(iMat);
G = G(iMat);
nu = nu(iMat);

eps_y = -nu/(1-nu)*(eps_x + eps_z); % required deformation along y to achieve plane
stress condition

eps = [eps_x eps_y eps_z 0 gamma_zx 0]';

% Constitutive law (linear, isotropic, elastic material):
C = E/(1+nu)/(1-2*nu)*[
        (1-nu) nu nu 0 0 0;
        nu (1-nu) nu 0 0 0;
        nu nu (1-nu) 0 0 0;
        0 0 0 (1-2*nu)/2 0 0;
        0 0 0 0 (1-2*nu)/2 0;
        0 0 0 0 0 (1-2*nu)/2];

sigma = C*eps;

stress_tensor = [sigma(1) 0 sigma(5); 0 sigma(2) 0; sigma(5) 0 sigma(3)];
principal_stresses = sort(eig(stress_tensor), 'descend');
sigma_I = principal_stresses(1);
sigma_II = principal_stresses(2);
sigma_III = principal_stresses(3);

% Von Mises
sigma_VM = sqrt(sigma_I^2 + sigma_II^2 + sigma_III^2 - sigma_I*sigma_II - sigma_II*
sigma_III - sigma_I*sigma_III);

%=====
% STATISTICS -- SCATTER PLOT + GAUSSIAN BELL (A vs B vs C)

```

```

%=====

close all
clear
clc
format long

% === FILE SETUP ===
fileList = {'experimental data A.txt', 'experimental data B.txt', 'experimental data C.txt'};
campaignNames = {'A', 'B', 'C'};

dataStruct = struct();

% === 1. LOAD DATA FROM EACH FILE ===
for i = 1:length(fileList)
    filename = fileList{i};
    opts = detectImportOptions(filename, 'NumHeaderLines', 6, 'VariableNamingRule', 'preserve');
    data = readtable(filename, opts);

    dataStruct(i).name = campaignNames{i};
    dataStruct(i).data = data;
end

% === 2. IDENTIFY NUMERIC COLUMNS (EXCEPT TIME) ===
columnNames = dataStruct(1).data.Properties.VariableNames;
numericCols = columnNames(~strcmp(columnNames, 'Time')); % exclude 'Time' column

% === 3. SCATTERPLOTS + STATISTICS + DISTRIBUTION PLOT FOR EACH PARAMETER ===
for c = 1:length(numericCols)
    param = numericCols{c};
    fprintf('\n=== Analysis: %s ===\n', param);

    allData = [];
    allGroups = [];

    % Create figure with two subplots: scatter + distribution
    figure('Name', ['Analysis - ', param], 'Position', [100, 100, 1000, 400]);

    % --- LEFT PLOT: SCATTERPLOT ---
    subplot(1, 2, 1);
    hold on;
    cmap = lines(length(dataStruct)); % different color for each dataset

    for i = 1:length(dataStruct)
        vec = dataStruct(i).data.(param);
        n = length(vec);
        x_jittered = i + 0.1 * (rand(n,1) - 0.5); % jittered x for better visibility

        scatter(x_jittered, vec, 10, 'filled', 'MarkerFaceColor', cmap(i,:));

        allData = [allData; vec];
        allGroups = [allGroups; repmat({dataStruct(i).name}, n, 1)];

        % Print stats
        fprintf('%s: mean = %.6f, std = %.6f, min = %.6f, max = %.6f\n', ...
            dataStruct(i).name, mean(vec), std(vec), min(vec), max(vec));
    end

    xlim([0.5, length(dataStruct) + 0.5]);
    xticks(1:length(dataStruct));

```

```
xticklabels(campaignNames);
ylabel(param);
title(['Scatterplot: ', param]);
grid on;

% --- RIGHT PLOT: NORMAL DISTRIBUTIONS (PDF) ---
subplot(1, 2, 2);
hold on;
for i = 1:length(dataStruct)
    vec = dataStruct(i).data.(param);
    mu = mean(vec);
    sigma = std(vec);

    % Create x-range for plotting normal PDF
    x_range = linspace(mu - 4*sigma, mu + 4*sigma, 200);
    y_pdf = normpdf(x_range, mu, sigma);

    plot(x_range, y_pdf, 'Color', cmap(i,:), 'LineWidth', 2, ...
        'DisplayName', ['Campaign ', dataStruct(i).name]);
end
title(['Gaussian Distribution: ', param]);
xlabel(param);
ylabel('Probability Density');
legend show;
grid on;
end

% === 4. TIME SERIES PLOTS ===
figure('Name', 'Time Series Plots');
for c = 1:length(numericCols)
    param = numericCols{c};
    subplot(ceil(length(numericCols)/2), 2, c);
    hold on;
    for i = 1:length(dataStruct)
        time = dataStruct(i).data.('Time');
        plot(time, dataStruct(i).data.(param), 'DisplayName', dataStruct(i).name);
    end
    title(['Time Series: ', param]);
    xlabel('Time [s]');
    ylabel(param);
    legend show;
    grid on;
end
```

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